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Author Full Name

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Programme Name

Supervisor:
Prof. Supervisor Name, PhD,
Position, Department,
Iscte – Instituto Universitário de Lisboa

Co-Supervisor:
Prof. Co-Supervisor Name, PhD,

Position, Department,
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Position, Department,
Institution

Month, Year

To my family.

Acknowledgments

I would like to thank my supervisor, **Prof. Supervisor Name**, for the guidance and support throughout this work.

Resumo

Este trabalho aborda o problema de . . .

Palavras-chave: palavra-chave 1, palavra-chave 2, palavra-chave 3.

Abstract

This thesis addresses the problem of …

Keywords: keyword 1, keyword 2, keyword 3.

List of Acronyms

Acronym	Definition
ACDC	Automated Cardiac Diagnosis Challenge
AHA	American Heart Association
ASSD	Average Symmetric Surface Distance
DSC	Dice Similarity Coefficient
ED	End-Diastole
EDT	Euclidean Distance Transform
ES	End-Systole
GNN	Graph Neural Network
HD95	95th-percentile Hausdorff Distance
ICC	Intraclass Correlation Coefficient
INR	Implicit Neural Representation
IoU	Intersection over Union
LV	Left Ventricle
LoA	Limits of Agreement
MAE	Mean Absolute Error
NOR	Normal cohort
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RMSE	Root Mean Squared Error
RV	Right Ventricle
SAX	Short-Axis
SDF	Signed Distance Function
SSM	Statistical Shape Model

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CHAPTER 1

Introduction

This chapter explains why the problem matters, describes the problem being addressed, states the research questions and objectives.

1.1. Motivation

The heart keeps blood moving through the body. The left ventricle (LV) has a central role because it pumps oxygen-rich blood into the aorta and from there to the rest of the body. Its size, shape, motion, and wall thickness can therefore provide important information about cardiac function. Changes in LV shape and wall thickness can also be associated with disease (de Marvao et al., 2014; Grossman et al., 1974).

Cardiac magnetic resonance imaging (MRI) allows the LV to be observed without an invasive procedure. Short-axis (SAX) images show the heart as a stack of slices from the base to the apex. These images make it possible to identify the inner and outer boundaries of the myocardium, which is the muscular wall of the heart. However, the slices alone do not provide a complete three-dimensional view. A 3D model can describe the full LV shape and support measurements in the regions between slices. Consistent descriptions of LV anatomy are useful in large imaging studies and automated analysis (Bai et al., 2015, 2018).

Myocardial wall thickness is one such measurement. A wall that is unusually thick, thin, or uneven can contain useful information about the state of the heart. In clinical images, wall thickness is often measured only on the acquired slices. A smooth 3D representation could provide a more complete and more easily visualised measurement across the LV. This thesis studies how sparse SAX observations can be used to reconstruct the LV in three dimensions and to measure myocardial wall thickness.

1.2. Problem Statement

SAX cardiac MRI is useful, but it is not a detailed 3D scan of the heart. The slices are separated along the long axis of the LV, leaving part of the shape unobserved between them. This makes 3D reconstruction difficult. Simply connecting the slices can create rough surfaces, gaps, or a myocardial wall that does not represent the anatomy well.

Two related surfaces must be reconstructed. The endocardium is the inner boundary of the LV cavity, and the epicardium is the outer boundary of the myocardium. The space between them must remain a valid wall everywhere. Public cardiac MRI datasets usually provide images and segmentation masks for each slice, rather than complete 3D LV meshes. Such meshes are needed to train and evaluate a surface-reconstruction model directly.

This work addresses the problem with a phase-conditioned implicit model trained using both synthetic LV shapes and meshes derived from real cardiac MRI segmentations. The model receives sparse contours as input and predicts continuous 3D LV surfaces. The outer surface is constrained to remain outside the inner surface, producing a myocardial wall with positive thickness. The reconstructed geometry is then used to study wall-thickness measurements and compare them with measurements from segmentation-derived geometry.

1.3. Research Questions

This work addresses three research questions. **RQ1** asks whether a model can reconstruct smooth, closed 3D LV surfaces from sparse SAX contours. **RQ2** asks whether the model can represent the endocardium and epicardium while maintaining a valid myocardial wall at both end-diastole and end-systole. **RQ3** asks how closely wall-thickness measurements from the reconstructed LV agree with measurements from segmentation-derived geometry.

1.4. Objectives

The first objective is to prepare synthetic and real cardiac MRI data in a shared contour-based format. The second objective is to develop a phase-conditioned signed-distance model that reconstructs the inner and outer LV surfaces from sparse SAX contours. A further objective is to ensure that the reconstructed surfaces are smooth and closed, with a myocardial wall of positive thickness. Finally, the reconstructed surfaces are evaluated against segmentation-derived meshes, and several methods for measuring local and regional wall thickness are compared.

1.5. Contributions

This thesis provides a data pipeline that converts synthetic LV shapes and real cardiac MRI segmentations into a shared representation of sparse SAX contours and 3D meshes. It also presents a phase-conditioned signed-distance model that predicts the endocardial and epicardial LV surfaces while preserving a positive wall between them. Finally, it evaluates the reconstructed geometry on data not used for training and compares local and regional wall-thickness methods on reconstructed and segmentation-derived LV geometry.

CHAPTER 2

Literature Review

This chapter presents the review methodology used to identify relevant literature and synthesises the related work on three-dimensional LV reconstruction from cardiac MRI and myocardial wall-thickness measurement.

2.1. Review Methodology

The literature search followed a structured approach to ensure broad coverage and relevance. This section describes the search strategy, selection criteria, and results.

2.1.1. Search strategy

The search was conducted across three major bibliographic databases: Scopus, PubMed, and IEEE Xplore, supplemented by Google Scholar for citation chaining. The temporal scope covered publications from 2000 to 2026, prioritising work from the last decade while retaining foundational methods essential for theoretical context.

Search terms were grouped into four thematic categories and combined using Boolean operators. The first category covered anatomy and imaging terms ("cardiac MRI", "left ventricle", "short-axis", "SAX", "myocardium"). The second addressed reconstruction ("3D reconstruction", "mesh reconstruction", "shape model", "implicit surface", "signed distance function"). The third targeted learning methods ("deep learning", "graph neural network", "point cloud", "neural representation"). The fourth focused on measurement ("wall thickness", "myocardial thickness", "Laplace equation", "transmural").

The refined query applied consistently across databases was:
("cardiac MRI" OR "left ventricle" OR "short-axis") AND ("3D reconstruction" OR "statistical shape model" OR "signed distance function" OR "graph neural network" OR "implicit representation") AND ("wall thickness" OR "mesh" OR "deep learning")

2.1.2. Inclusion and exclusion criteria

Studies were included if they: (i) addressed three-dimensional reconstruction or shape modelling of the LV, (ii) proposed or evaluated methods for myocardial wall-thickness measurement, (iii) developed geometric deep learning techniques applicable to anatomical meshes, or (iv) introduced benchmark cardiac MRI datasets used for LV analysis.

Studies were excluded if they: (i) focused exclusively on segmentation without reconstruction or shape analysis, (ii) addressed organs other than the heart without methodological generalisability, (iii) lacked peer review or sufficient methodological description, or (iv) were published before 2000 unless they constituted foundational work directly relevant to the methods reviewed.

2.1.3. Results

The initial search retrieved 142 records across the three databases. After removing 31 duplicates, 111 unique records were screened by title and abstract. Of these, 48 were selected for full-text assessment. Following detailed review against the inclusion criteria, 28 studies were retained. An additional 7 studies were identified through backward citation searching, yielding a final corpus of 35 publications.

The selected studies span five thematic areas: cardiac MRI acquisition and segmentation, SSMs, INRs, GNNs for mesh processing, and wall-thickness measurement. The following section synthesises the key findings from this corpus.

2.2. Related Work

This section synthesises the reviewed literature, organised by thematic area.

2.2.1. Cardiac MRI segmentation and datasets

Short-axis (SAX) cine MRI is the standard protocol for left-ventricle (LV) evaluation, producing 2D slices perpendicular to the long axis with high in-plane resolution (1–2 mm) but large inter-slice gaps (6–10 mm) (Cerqueira et al., 2002; Petitjean & Dacher, 2011). This anisotropic acquisition creates the fundamental challenge that motivates 3D reconstruction: the ventricle must be inferred volumetrically from sparse planar observations. Petitjean and Dacher (2011) reviewed over 100 segmentation methods and concluded that the inter-slice gap remains the single largest source of error in volumetric cardiac analysis, with apex and base delineation accounting for the majority of inter-observer variability.

The AHA 17-segment model, introduced by Cerqueira et al. (2002), standardised the partition of the myocardium into basal, mid-cavity, and apical regions. It is widely used for reporting

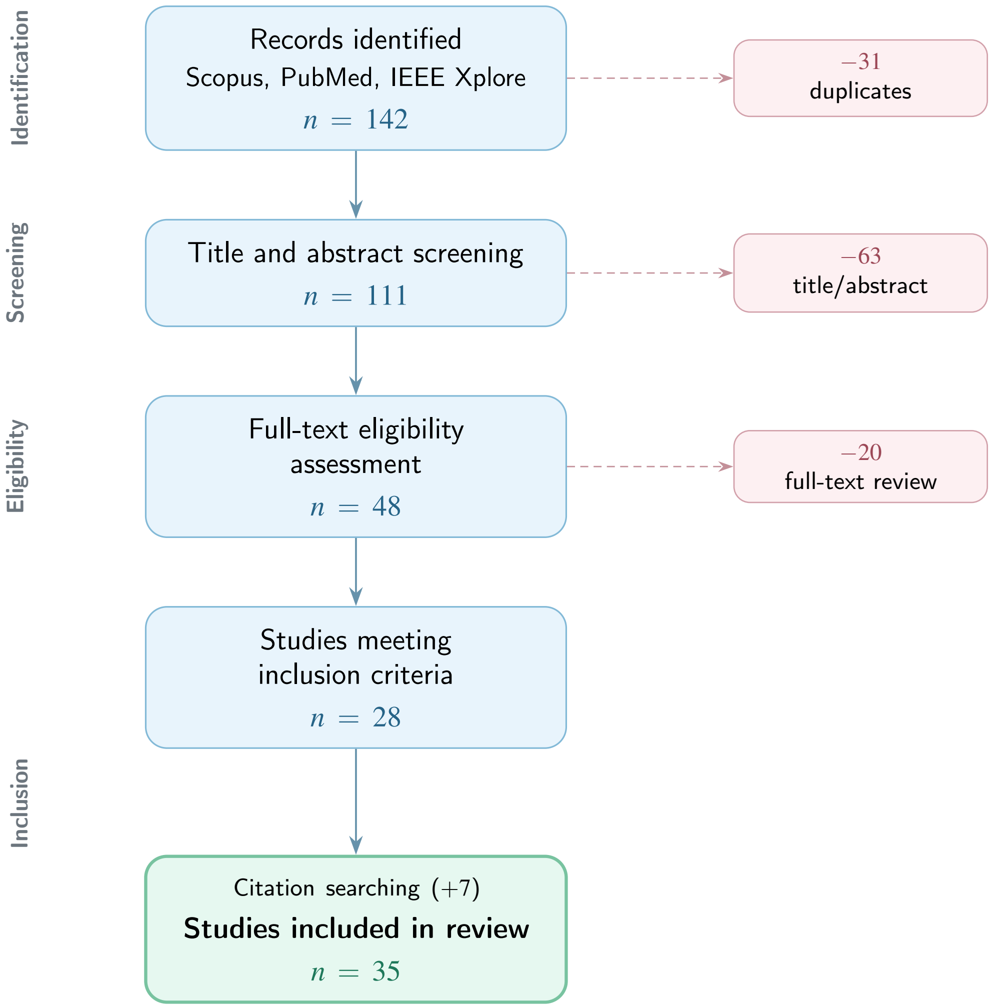

Figure 2.1. PRISMA flow diagram summarising the literature selection process.

wall motion, perfusion, and thickness, and serves as the reference frame for all regional analysis discussed in subsequent sections.

Segmentation of endocardial and epicardial contours is a prerequisite for any downstream 3D analysis. Ronneberger et al. (2015) proposed U-Net, an encoder–decoder architecture with skip connections that was originally designed for neuronal structure segmentation and was later widely used for cardiac MRI analysis. Its encoder–decoder design combines image context with fine spatial detail, which is useful when extracting endocardial and epicardial contours from SAX slices.

The Automated Cardiac Diagnosis Challenge (ACDC), organised by Bernard et al. (2018), provided a standardised benchmark for cardiac segmentation and diagnosis. The dataset comprises 100 patients distributed across five clinical groups: normal, dilated cardiomyopathy (DCM), hypertrophic cardiomyopathy (HCM), abnormal right ventricle, and myocardial infarction with altered LV ejection fraction. Each case includes expert delineations at ED and ES. The challenge reported that top-performing deep learning methods achieved mean Dice scores exceeding 0.93 for the LV cavity and 0.89 for the myocardium, demonstrating that automated segmentation had reached expert-level performance for standard acquisitions. The images and their expert labels remain openly available for research at no cost, after a free registration and on the condition that the challenge paper is cited.

Isensee et al. (2021) introduced nnU-Net, a self-configuring framework that automatically adapts network topology, patch size, preprocessing, and post-processing to each dataset. Applied to the ACDC benchmark, nnU-Net matched or exceeded all prior challenge submissions without any manual architecture design. The framework has since become the standard baseline against which new cardiac segmentation methods are evaluated.

The Multi-Centre, Multi-Vendor and Multi-Disease (M&Ms) challenge, described by Campello et al. (2021), extended evaluation to the clinically critical question of domain generalisation. The dataset includes 375 CMR studies acquired on scanners from four different vendors (Siemens, General Electric, Philips, and Canon) across six clinical centres. The challenge demonstrated that while methods trained on a single vendor achieved high accuracy on that vendor, performance dropped significantly when tested on unseen vendors — with Dice reductions of 5–15 percentage points in some cases. This finding motivated the development of domain-adaptation techniques and highlighted the importance of multi-source training data. As with ACDC, the data are distributed free of charge through the challenge website after registration.

For population-level analysis, the UK Biobank (Sudlow et al., 2015) provides the largest available resource, with cardiac MRI acquired for over 40 000 participants using a standardised 1.5T protocol. Bai et al. (2018) developed a fully convolutional pipeline to segment this cohort at scale, producing automated delineations that, after quality control, serve as the foundation for large-scale shape modelling and epidemiological studies. Unlike the challenge datasets described above, the UK Biobank is not openly downloadable. Access is granted through an approved application and is subject to a tiered access fee, with imaging data placed in the highest tier.

2.2.2. Statistical shape models

Statistical Shape Models (SSMs) encode anatomical variability by aligning a training population to point-to-point correspondence and applying PCA to extract the dominant modes of variation (Cootes et al., 1995; Heimann & Meinzer, 2009). Cootes et al. (1995) introduced Point Distribution Models, where a training set of shapes — each represented by k corresponding landmarks — is rigidly aligned via Procrustes analysis, averaged into a mean shape $\bar{x}$, and decomposed via PCA into principal modes Φ. A new shape is generated as $x = \bar{x} + \Phi b$, where b

are shape parameters constrained within $\pm 3\sigma$ of each eigenvalue to remain anatomically plausible. In their experiments on hand and face contours, Cootes et al. (1995) showed that the first 5–10 modes captured over 95% of population variance, demonstrating the efficiency of the linear representation.

Heimann and Meinzer (2009) surveyed over 100 studies applying SSMs to medical image segmentation and identified the cardiac domain as particularly well-suited because of the regularity of ventricular geometry and the availability of large training cohorts. Their review highlighted that SSM accuracy depends critically on three factors: the quality of inter-subject correspondence, the size and diversity of the training set, and the alignment method used.

Frangi et al. (2002) pioneered automatic construction of multi-object cardiac SSMs, proposing a non-rigid registration framework that establishes dense correspondence across subjects without manual landmarks. Applied to biventricular models, their approach captured coupled shape variation between the left and right ventricles, demonstrating that multi-structure models outperform single-structure ones for segmentation-by-fitting tasks.

Bai et al. (2015) constructed the most comprehensive cardiac atlas to date, using over 1000 high-resolution MRI scans from the UK Digital Heart Project. Their biventricular model captures both end-diastolic shape and wall motion through the cardiac cycle. The authors reported that the first 50 PCA modes explain approximately 90% of shape variance, and demonstrated that the atlas enables automatic segmentation of new subjects by fitting the model to image evidence with mean surface-to-surface errors below 1.5 mm. An important practical point is that the derived shape model is released openly and free of charge, while the underlying patient images are not: only the mean shape, the principal components, and the corresponding variances are distributed. A shape model of this kind can therefore be reused without the access restrictions that apply to the imaging cohort it was built from.

de Marvao et al. (2014) applied the same UK Digital Heart Project atlas to study myocardial hypertrophy and showed that atlas-based shape analysis detects subtle remodelling patterns that volumetric indices alone (such as LV mass) miss. Their study demonstrated substantially improved statistical power for detecting hypertrophic phenotypes when using shape modes compared to traditional clinical measurements, requiring fewer subjects to achieve equivalent sensitivity.

Medrano-Gracia et al. (2014) used a multi-ethnic cohort of over 1900 subjects to characterise normal LV shape variation. They reported that the dominant modes correspond to overall ventricular size, followed by eccentricity, apical shape, and regional wall curvature — confirming that cardiac shape cannot be adequately described by a single size parameter. Suinesiaputra et al. (2018) organised an international challenge to classify myocardial infarction from SSM-derived shape features, showing that machine learning classifiers trained on the first 20 shape modes achieved AUC values above 0.85 for distinguishing infarcted from healthy ventricles.

SSMs serve dual roles in reconstruction pipelines. First, they act as *geometric priors* that regularise shape recovery from sparse observations, constraining the solution to lie on or near the learned manifold. Second, they function as *synthetic data generators*: by sampling the latent space, one can produce thousands of anatomically plausible meshes for training data-hungry deep learning models without additional manual annotation (Heimann & Meinzer, 2009; Khalafvand et al., 2018). Khalafvand et al. (2018) demonstrated this second role by generating SSM-sampled LV geometries to train computational fluid dynamics simulations, showing that the synthetic population reproduced the haemodynamic statistics of real subjects.

2.2.3. Deep learning models for 3D shape reconstruction

Classical reconstruction methods interpolate between segmented SAX slices using splines or deformable surfaces, but cannot recover geometric detail between slices, particularly near the apex and base (Heimann & Meinzer, 2009; Petitjean & Dacher, 2011). Petitjean and Dacher (2011) noted that linear interpolation between contours produces visible staircase artefacts and fails to reproduce the smooth taper of the LV apex, while parametric models (prolate spheroids, B-spline surfaces) impose smoothness but cannot express patient-specific morphological detail. Modern approaches replace interpolation with learned shape recovery, falling into two broad families: template deformation and implicit field prediction.

Template deformation methods maintain a fixed mesh topology and learn per-vertex displacements conditioned on input observations. Wang et al. (2018) proposed Pixel2Mesh, which takes a single RGB image as input and iteratively refines an ellipsoidal template mesh through three stages of graph convolutional layers. At each stage, vertex features are enriched with image features sampled via learned projections, and the mesh is refined through unpooling. The authors reported Chamfer distances below 0.5 on the ShapeNet benchmark, demonstrating that graph-based deformation can recover complex topology from minimal input. Choi et al. (2020) extended this paradigm with Pose2Mesh, which recovers full 3D human body meshes from sparse 2D pose keypoints. Their cascaded coarse-to-fine architecture first predicts a coarse mesh from joint positions and then refines vertex locations with finer graph convolutions, achieving state-of-the-art results on the Human3.6M benchmark. The architectural principle, recovering dense 3D geometry from sparse 2D observations, is directly analogous to the cardiac problem of inferring a full LV mesh from a handful of 2D SAX contours.

In the cardiac domain, template deformation preserves the topological consistency required for cross-subject comparison and downstream analysis such as wall-thickness computation (Bai et al., 2015; Suinesiaputra et al., 2014). Beetz et al. (2021) applied point-cloud variational autoencoders to biventricular shapes from the UK Biobank, learning a compact latent space that enables generation of subpopulation-specific anatomy models. Their decoder produces meshes with fixed vertex correspondence, making population statistics directly computable from the generated samples.

Implicit Neural Representations (INRs) take a different approach. Rather than deforming a template, they encode geometry as a continuous function whose zero-level set defines the surface. Park et al. (2019) introduced DeepSDF, which parameterises the signed distance function (SDF) of an entire shape class with a single neural network conditioned on a per-shape latent code. Given partial or noisy point-cloud observations, their auto-decoder architecture optimises the latent code at test time to find the shape in the learned space that best explains the input. On the ShapeNet benchmark, DeepSDF achieved reconstruction quality comparable to voxel-based methods while using an order of magnitude fewer parameters, and naturally supports shape interpolation in latent space. Mescheder et al. (2019) proposed Occupancy Networks, which predict binary inside/outside occupancy rather than signed distance. Their encoder processes a point cloud or image to produce a latent vector, and the decoder evaluates occupancy at arbitrary 3D query points. Both DeepSDF and Occupancy Networks produce resolution-independent surfaces extracted via the Marching Cubes algorithm (Lorensen & Cline, 1987), which generates a watertight triangulated mesh from the zero-crossing of the implicit field on a regular grid.

Two techniques are critical for applying INRs to thin cardiac structures where high-frequency geometric detail must be preserved. Tancik et al. (2020) demonstrated that standard MLPs exhibit spectral bias — they preferentially learn low-frequency components — and proposed mapping input coordinates through random Fourier features before the network. In their experiments on 2D image regression and 3D shape reconstruction, Fourier encoding substantially reduced reconstruction error compared to raw coordinate input, with the improvement most pronounced for high-frequency details such as thin structures and sharp edges. Gropp et al. (2020) introduced Implicit Geometric Regularisation, showing that enforcing the eikonal constraint ($\|\nabla f\| = 1$) as a loss term during training produces valid distance fields without requiring ground-truth SDF values at every point. Their approach enables training from raw point clouds alone, which is particularly relevant when dense SDF supervision is expensive to compute.

For encoding unordered input observations such as SAX contour points, the PointNet architecture (Qi et al., 2017) provides a permutation-invariant encoder via shared MLPs followed by symmetric max-pooling. Qi et al. (2017) demonstrated that this simple design achieves classification accuracy of 89.2% on the ModelNet40 benchmark and produces compact global descriptors that capture the essential geometry of the input point set regardless of its size or ordering. In cardiac applications, each SAX slice contributes a set of contour points that can be processed by a PointNet-style encoder to produce a latent representation of the observed geometry.

2.2.4. Wall-thickness measurement techniques

Myocardial wall thickness is a primary clinical biomarker for cardiac remodelling. Grossman et al. (1974) demonstrated that wall thickness is a principal determinant of left-ventricular diastolic stiffness, with hypertrophied ventricles exhibiting markedly elevated stiffness proportional to end-diastolic wall thickness. More recently, Huelnhagen et al. (2017) showed at 7.0T MRI that wall thickness correlates with tissue transverse relaxation properties, confirming its value as a non-invasive surrogate for myocardial tissue characterisation. The AHA 17-segment model (Cerqueira et al., 2002) provides the standard framework for reporting regional thickness values.

The accuracy of wall-thickness measurement depends critically on the quality of the underlying surface reconstruction, as any smoothing or geometric artefact propagates directly into the thickness estimate. The literature describes five families of measurement methods, each with distinct trade-offs.

The simplest approach uses nearest-neighbour distance: for each endocardial vertex, a KD-tree identifies the closest epicardial point, yielding a measurement in $O(N \log M)$ time. While computationally trivial (under 1 ms for typical cardiac meshes), this method provides only a lower bound on the true transmural thickness because it ignores the anatomical direction of measurement — the nearest point may lie tangentially rather than across the wall.

Ray-based methods address this limitation by casting a ray from each endocardial vertex along its surface normal until it intersects the epicardium. The intersection is computed via the Möller–Trumbore algorithm, accelerated with a Bounding Volume Hierarchy (BVH) to $O(\log F)$ per query. This approach measures transmurally but is sensitive to normal noise: a slightly rotated normal can produce a spurious long or short ray. The Shape Diameter Function variant improves robustness by casting a cone of K rays (typically $K{=}7$) at a half-angle of 30°

around the normal and reporting the median intersection distance, effectively filtering outlier hits caused by local mesh irregularities.

Volumetric methods operate on voxelised representations. The Euclidean Distance Transform (EDT) computes, for each myocardial voxel, the distance to the nearest boundary, and the sum of the endocardial and epicardial distance fields approximates local thickness ($t = D_{\text{endo}} + D_{\text{epi}}$). This estimate is exact for planar parallel boundaries and overestimates by approximately 5–10% for the curvatures typical of the LV mid-wall. The method is computationally efficient ($O(V)$ via the Saito–Toriwaki algorithm) but limited by voxel resolution.

PDE-based methods are widely used for transmural thickness estimation. Jones et al. (2000) solved Laplace's equation between the two tissue boundaries and used the resulting non-intersecting field lines to establish corresponding paths. Thickness is obtained from the path length between the boundaries along these streamlines. Yezzi and Prince (2003) formulated an Eulerian PDE method that computes distances propagated from both boundaries along the Laplacian field. Their sum gives the local tissue thickness.

Finally, mesh-based regularised correspondence methods combine an initial geometric estimate (normal projection or nearest neighbour) with iterative Laplacian smoothing on the mesh connectivity graph. At each iteration, each vertex's thickness is updated as a weighted average of its neighbours, anchored to the initial estimate. This produces spatially smooth thickness maps suitable for bullseye visualisation, at the cost of potentially attenuating real local thickness variation if the smoothing parameter is set too aggressively.

2.2.5. Summary and research gap

The reviewed literature shows that each component of the LV analysis pipeline is well-studied in isolation: deep learning gives accurate slice-wise segmentation, SSMs encode population-level shape priors, implicit representations provide continuous surface modelling, and multiple wall-thickness algorithms exist. To the best of our knowledge, however, limited work integrates these components into a single pipeline for the specific setting addressed in this thesis. The combination that appears to be missing is a method that encodes sparse SAX contour points, decodes paired endocardial and epicardial surfaces as implicit signed distance fields, conditions the reconstruction on cardiac phase, constrains the wall offset to stay positive, and trains jointly on real clinical data and SSM-derived synthetic samples. This is useful because the wall can then be measured everywhere on a continuous surface instead of only on the acquired slices. This gap motivates the approach described in the remaining chapters.

CHAPTER 3

Methodology

This chapter describes the methodology used to reconstruct 3D LV geometry from sparse SAX contours and to measure myocardial wall thickness on the reconstructed model. The proposed pipeline combines synthetic and real cardiac MRI-derived data, a phase-conditioned INR, and a systematic comparison of local wall-thickness algorithms.

3.1. Overview

The complete thesis workflow can be read as a progression from sparse clinical observations to continuous three-dimensional geometry and then to quantitative wall-thickness summaries. It starts from segmented SAX cardiac MRI stacks or synthetic SSM meshes. In both cases, the available geometry is converted into a common contour-based representation composed of endocardial and epicardial points sampled across SAX slices. These points are passed to a PointNet-style encoder that produces a compact latent code for the observed LV geometry. A signed-distance implicit decoder then predicts two coupled surfaces: the endocardium and the epicardium. The epicardial field is parameterised as a monotone offset from the endocardial field, which guarantees positive wall thickness by construction. The resulting surfaces are finally used to compute local thickness maps and regional AHA-17 summaries.

The methodological design is organised around four objectives. First, the input representation must be compatible with both synthetic and real cases. Second, the model must reconstruct smooth watertight LV surfaces from sparse 2D observations. Third, the decoder must produce a valid myocardial wall everywhere, not only at the observed slices. Fourth, the reconstructed geometry must support local wall-thickness measurements that can be visualised as surface colour maps and regional AHA-17 bullseye plots.

The chapter is therefore organised as a technical pipeline rather than as a single model description. Section 3.2 explains how synthetic and real data were converted into a shared cache representation. Section 3.3 describes the model encoder and decoder, with particular attention to the monotone-epi parameterisation used to guarantee positive wall thickness. Section 3.4 presents the training and inference procedure, and Section 3.5 describes the wall-thickness algorithms used to evaluate local measurements on the reconstructed geometry.

Figure 3.1 summarises this end-to-end workflow and should be interpreted as an overview of the thesis methodology rather than as a figure belonging only to data preparation. The first panels illustrate the sparse SAX observation and contour extraction, the mesh panel represents the reconstructed LV surface domain, and the lower panels show how the reconstructed geometry is converted into local and regional wall-thickness measurements. In this preview, panels (a) and (b) are drawn from the ACDC Challenge dataset, whereas panels (c), (d), and (e) are drawn from the UK Digital Heart Project LV SSM pipeline.

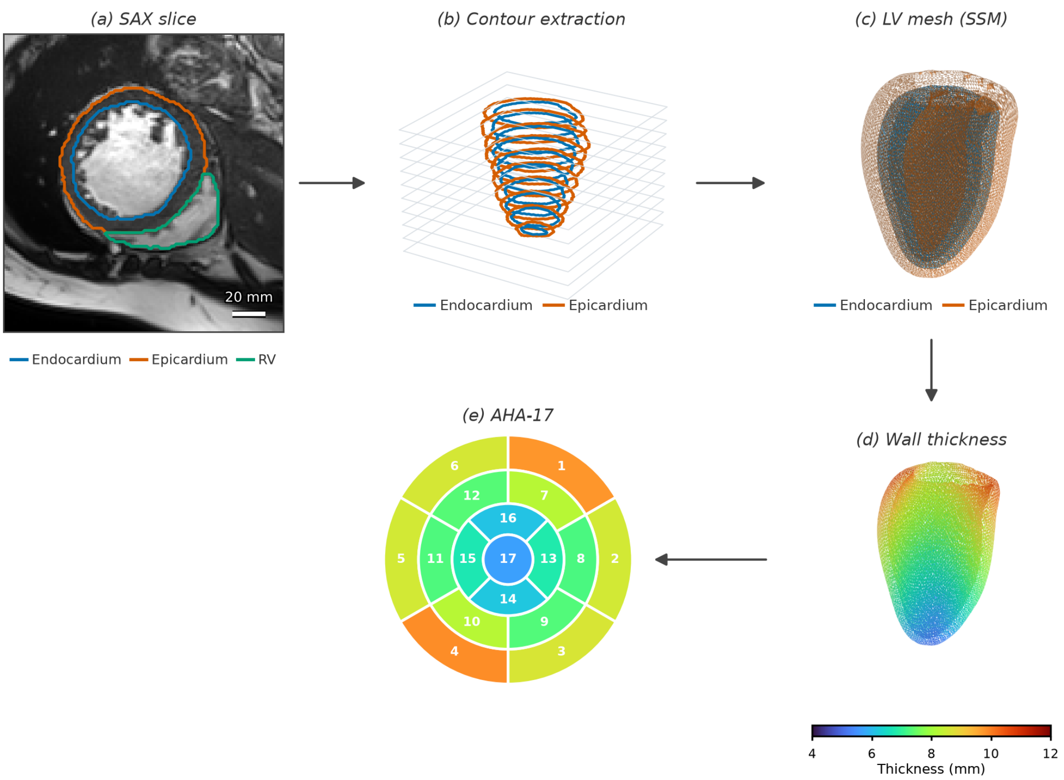

Figure 3.1. Preview of the end-to-end methodology from sparse SAX observations to reconstructed LV geometry and wall-thickness outputs.

3.2. Data Preparation

A fundamental challenge in training the model is the absence of a paired dataset that provides both sparse SAX contour observations *and* corresponding validated three-dimensional LV meshes. Clinical cardiac MRI datasets supply segmentation masks and image volumes, but they do not include the watertight endocardial and epicardial surfaces that an SDF decoder requires as supervision targets. This missing ground-truth problem is the starting point for the entire data-preparation strategy.

To address this issue, a two-stage approach was used. In the first stage, 1300 synthetic ED meshes were generated from the UK Digital Heart Project SSM of the LV, providing anatomically plausible three-dimensional geometry with controlled shape variation. The model was pre-trained on this synthetic corpus, which established a geometric prior for LV shape variability. In the second stage, the training corpus was extended with meshes reconstructed directly from real clinical segmentations: ED and ES frames from the ACDC and original M&Ms datasets. This fine-tuning stage introduced patient-specific anatomy, acquisition variability, and phase diversity that the SSM alone cannot provide.

All three data streams were converted into the same cache schema. The shared format allows the pre-training and fine-tuning stages to use the same training loop and loss function without any dataset-specific logic.

3.2.1. The missing ground-truth problem and motivation for SSM

The limitation outlined above can be stated more precisely as a supervision-gap problem: the model requires paired samples of sparse SAX contours and reliable three-dimensional LV meshes, while public cine MRI benchmarks provide mainly slice-wise labels. In the ACDC (Bernard et al., 2018) and original M&Ms (Campello et al., 2021) datasets, the available annotations are segmentation masks rather than watertight endocardial and epicardial surfaces suitable for direct SDF supervision.

Two complementary strategies were therefore adopted to create the missing supervision. First, an SSM was used to generate a large number of synthetic ED meshes that are anatomically valid by construction. This solved the ground-truth problem for the pre-training stage at the cost of not capturing real segmentation artefacts or scanner-specific anatomy. Second, real ED and ES segmentation masks were converted into watertight three-dimensional meshes through a controlled preprocessing pipeline. This provided the patient-specific geometry needed for fine-tuning, at the cost of requiring careful mesh reconstruction from inherently discrete voxel data.

The SSM pre-training approach is motivated by the observation that learning the geometry of the LV, rather than fitting individual segmentations, requires exposure to the full range of normal and pathological LV shapes. The SSM encodes this range compactly as a mean shape and a set of principal modes of variation, and sampling from its latent space produces diverse but anatomically constrained meshes. The real-data fine-tuning stage then ensures that the model generalises to the segmentation quality and acquisition conditions present in the evaluation datasets.

3.2.2. Statistical Shape Model of the left ventricle

An SSM represents the variability of an anatomical structure across a population by applying principal component analysis (PCA) to a set of aligned training meshes (Heimann & Meinzer, 2009). Each mesh is expressed as a vector of vertex coordinates; the mean of all training shapes forms the mean shape $\bar{X}$, and the eigenvectors of the covariance matrix of the centred shapes form the principal modes. Any shape in the population can be approximated as a linear combination of these modes, and new shapes that follow the observed statistics can be generated by sampling the latent coefficient vector.

This thesis uses the UK Digital Heart Project LV SSM reported by Bai et al. (2015), built from more than 1,000 high-resolution cardiac MR images of healthy subjects. The model provides a mean endocardial surface of 22 043 vertices together with 100 principal modes of variation stored as a $66\,129 \times 100$ matrix of mode vectors and a corresponding vector of 100 eigenvalues λ_i. The standard deviations $\sigma_i = \sqrt{\lambda_i}$ quantify the magnitude of each mode. A new endocardial mesh is generated by

$$X(b) = \bar{X} + \Phi b, \tag{3.1}$$

where $\bar{X}$ is the mean shape flattened to a vector, Φ is the matrix of mode vectors, and $b = [b_1, \ldots, b_{100}]^\top$ is the sampled coefficient vector.

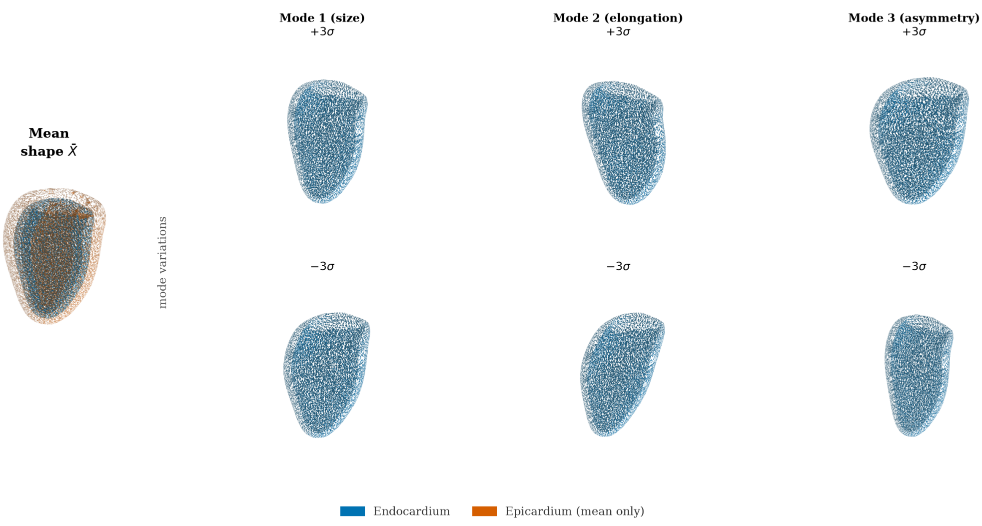

Figure 3.2. UK Digital Heart Project LV SSM mean shape and first three PCA modes at $\pm 3\sigma$.

The first few modes capture the largest sources of shape variation in the population; mode 1 primarily controls overall LV size, mode 2 encodes elongation along the long axis, and higher modes represent progressively more localised deformations, as shown in Figure 3.2.

3.2.3. Synthetic end-diastolic dataset (SSM pre-training)

The synthetic ED dataset was built by sampling 1300 meshes from the SSM and converting each mesh into a training-compatible sample. The generation pipeline proceeds in five steps, illustrated in Figure 3.3.

Step 1: Latent coefficient sampling. Each candidate coefficient vector b was drawn by sampling each component as $b_i \sim \mathcal{N}(0, \sigma_i^2)$ and then applying two acceptance criteria:

$$\left|\frac{b_i}{\sigma_i}\right| \leq 3, \qquad \sum_i \left(\frac{b_i}{\sigma_i}\right)^2 \leq \chi^2_{0.99,d}, \tag{3.2}$$

where σ_i is the standard deviation of mode i, d is the number of active modes, and $\chi^2_{0.99,d}$ is the 99th percentile of the chi-squared distribution with d degrees of freedom. The per-mode clipping prevents individual modes from dominating the shape, and the Mahalanobis bound prevents the combined deformation from falling outside the plausible population distribution. Accepted samples are scattered within a bounded ellipsoid in latent space, as shown in panel (c) of Figure 3.3.

Step 2: Mesh generation. Each accepted coefficient vector b is substituted into Equation (3.1) to produce a deformed endocardial vertex array, which is then triangulated using the face connectivity of the mean-shape mesh. The resulting triangular surface is watertight and oriented by construction.

Step 3: Quality filtering. Before a generated mesh is accepted into the training set, it is evaluated against four proxy metrics: enclosed volume (mL), surface area (cm^2), sphericity index, and plausibility of the apical and basal normal directions. Samples that fall outside calibrated bounds for any of these metrics are discarded. Periodic exact audits using VTK mass-properties computation were also performed every 40 samples to verify the proxy metric calibration.

Step 4: Synthetic epicardium. The SSM provides only the endocardial surface. The epicardial surface was synthesised by offsetting each endocardial vertex p_i along its outward normal n_i by a spatially varying distance d_i,

$$q_i = p_i + d_i n_i, \tag{3.3}$$

where d_i was set to approximately 10 mm near the base and 5 mm near the apex, with additive Gaussian noise of standard deviation 0.5 mm to avoid perfectly uniform geometry. This synthetic epicardium is a geometric pre-training prior only. It is not intended to reproduce clinical wall thickness, and it is never used as a reference for any wall-thickness result. Its purpose is to give the decoder a consistent outer surface during pre-training, so that the model learns the general shape relation between an inner and an outer LV surface before it sees real data.

Step 5: SAX contour extraction and volumetric supervision points. Each pair of endocardial and epicardial surfaces was sliced at 10 evenly spaced axial levels to produce sparse SAX contour rings, matching the acquisition pattern of real cine MRI. For each case, 2048 query points were sampled near the surfaces and in the surrounding volume, and signed-distance targets or inside/outside labels were computed. These query points form the dense volumetric supervision used by the SDF loss terms during training.

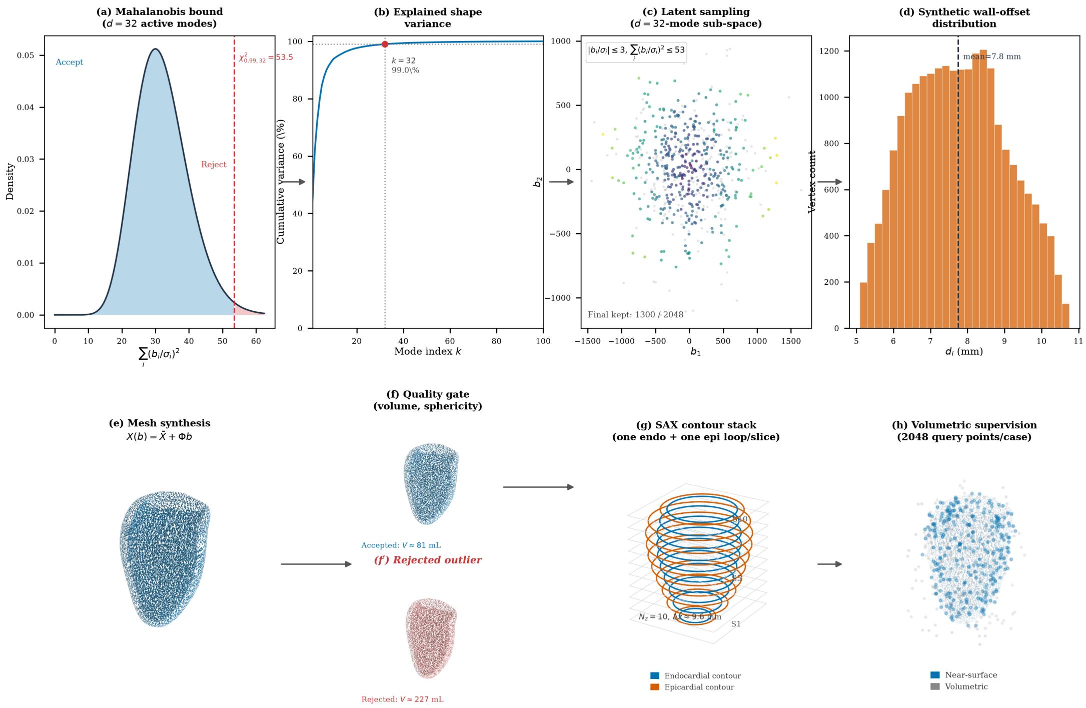

Figure 3.3. Synthetic ED dataset generation from SSM sampling to geometric supervision. Top row: statistical constraints; bottom row: geometric construction and supervision targets.

Figure 3.3 summarises this process in two rows. The top row provides the statistical view of sampling: panel (a) shows the Mahalanobis-statistic density and the $\chi^2_{0.99,32}$ acceptance threshold, panel (b) shows that 32 active modes capture 99% cumulative shape variance, panel (c) shows accepted latent samples in the first two dimensions, and panel (d) shows the synthetic epicardial-offset distribution from Equation (3.3). The bottom row provides the geometric view: panel (e) shows mesh synthesis via Equation (3.1), panel (f) contrasts an accepted shape with a rejected outlier, panel (g) shows the 10 SAX contour levels, and panel (h) shows the 2048 near-surface and volumetric query points used for decoder supervision.

These fixed SSM-sampling and supervision settings were kept constant across all synthetic ED cases to ensure reproducible pre-training before fine-tuning on real ED and ES data.

3.2.4. Real ED and ES labels: from SAX segmentation to 3D supervision

SSM-derived meshes alone cannot capture the full diversity of clinical acquisition conditions: scanner differences, segmentation quality, image resolution, and the distinct geometry of the end-systolic phase. A second data stream was therefore created by reconstructing watertight three-dimensional LV meshes directly from the segmentation masks provided in the ACDC (Bernard et al., 2018) and original M&Ms (Campello et al., 2021) datasets, taking both the ED and ES frames so that the model is exposed to both cardiac phases.

These real ED and ES caches are used in the second optimisation stage, after the initial SSM-based training. The reason for this fine-tuning stage is that the synthetic corpus teaches a strong global LV-shape prior, but it does not fully reproduce clinical variability in contour quality, inter-dataset label style, and phase-specific contraction patterns. Fine-tuning on real ED and ES therefore reduces domain shift between synthetic supervision and the target clinical setting while preserving the geometric regularity learned from the SSM.

Operationally, the fine-tuning stage keeps the same model architecture and the same multi-term loss from Section 3.4, but replaces the synthetic-only stream with real-derived cache samples from ED and ES frames. Because the real caches follow the same schema (contours, tissue labels, phase label, mesh supervision, and query points), the training loop can continue from the pre-trained checkpoint without changing model interfaces. The phase label then provides the conditioning signal that allows one shared model to adapt from diastolic to systolic geometry during continued training.

Before describing how these meshes are built, it is useful to look at the raw clinical input, because it makes clear why a reconstruction step is needed at all. Figure 3.4 shows representative SAX slices from the ACDC dataset (Bernard et al., 2018), with the LV cavity, myocardium, and right-ventricular contours overlaid. Each frame is only a two-dimensional cross-section of the heart, and the segmentation is defined one slice at a time. Panel (b) of Figure 3.4 stacks the segmented ED slices at their physical through-plane positions and exposes the central difficulty of the problem: a clinical acquisition provides only a small number of parallel SAX slices separated by several millimetres, so the observed geometry is a set of sparse contour rings rather than a continuous surface.

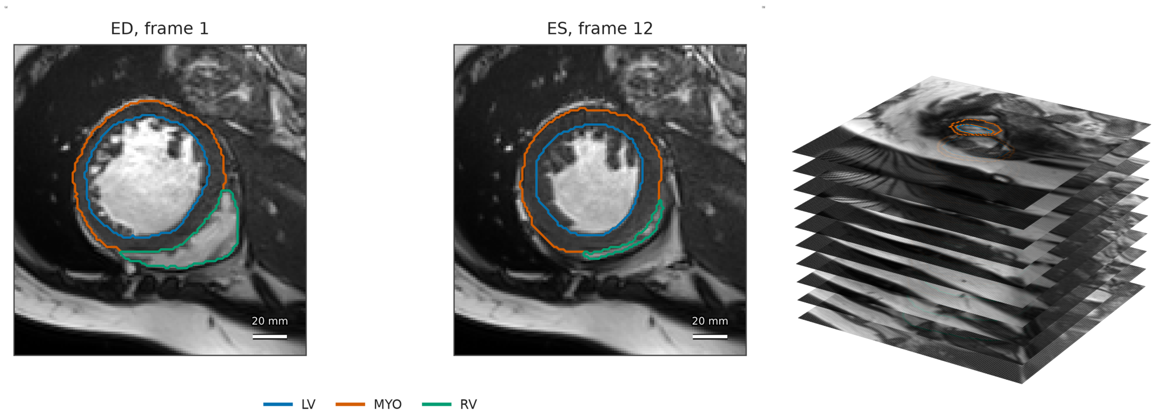

Figure 3.4. Clinical-input view used in the real-data pipeline: (a) representative ACDC ED/ES SAX slices with labels and (b) the corresponding sparse ED SAX stack in physical three-dimensional spacing.

The reconstruction pipeline closes this gap by converting each sparse label stack into a derived, watertight three-dimensional mesh. These meshes are *derived supervision*, not manually annotated 3D reference geometry: their geometry is determined by the observed labels and by the explicitly stated resampling, smoothing, and capping operations. This distinction is important when interpreting the later reconstruction results. An earlier ES pipeline attempted to fit the SSM to systolic contours; this was abandoned because forcing a diastolic statistical prior onto systolic geometry produced anatomically unrealistic surfaces with collapsed walls. The direct reconstruction approach instead retains the patient-specific geometry present in the ED or ES labels. The two parallel endocardial and epicardial paths are shown in Figure 3.5, and the four stages of the conversion are described below. Table A.2 of Appendix A lists the individual operations in order.

From label volume to clean binary masks. Each segmentation volume is read from NIfTI format and reoriented to the RAS+ convention using its affine, which places the voxel axes in a consistent patient frame while preserving the physical spacing. This step does not define the cardiac long axis. The acquired SAX stack supplies the through-plane direction, and base and apex are identified later from the cross-sectional areas at the two ends of the stack. The ACDC and original M&Ms datasets use compatible but not identical label encodings for the LV cavity, the myocardium, and the right ventricle, so each source is first mapped to a common convention. Two binary masks are then formed. The endocardial mask is the LV cavity label, and the epicardial mask is the union of the cavity and the myocardium:

$$M_{\text{endo}} = \mathbb{1}[L = L_{\text{LV}}], \qquad M_{\text{epi}} = \mathbb{1}[L \in \{L_{\text{LV}}, L_{\text{MYO}}\}], \tag{3.4}$$

where L is the label image, L_{LV} is the LV cavity label, and L_{MYO} is the myocardium label. The volume is then resampled to 1.5 mm isotropic spacing with nearest-neighbour interpolation, which avoids blending one label into another. Within each slice, the largest connected component of each label is kept and any holes are filled, and the largest three-dimensional component is kept afterwards. This removes the small disconnected fragments that often appear near the first and last slices of the stack. The endocardial mask is finally merged into the epicardial one so that $M_{\mathrm{endo}} \subseteq M_{\mathrm{epi}}$ holds everywhere.

From masks to a smooth closed surface. Applying marching cubes directly to voxel labels produces a visible staircase, so the binary mask is first smoothed with a Gaussian filter at a scale of 1.5 mm and the surface is taken at its 0.5 isosurface (Lorensen & Cline, 1987). Panels (d) and (d′) of Figure 3.5 show the unsmoothed voxel surface for the endocardial and epicardial masks, which makes this artefact explicit. Taubin λ/μ smoothing (Taubin, 1995) (25 iterations, $\lambda = 0.53$, $\mu = -0.55$) then removes most of the remaining staircase while approximately preserving the enclosed volume. Because the labels stop at the valve plane, marching cubes closes the top of the ventricle with a rounded dome that does not exist in the anatomy. The upper 6% of the apex-to-base extent is therefore cut away, the opening is closed with a flat cap perpendicular to the long axis, and three iterations of cotangent-Laplacian fairing smooth the join. Quadric-error decimation (Garland & Heckbert, 1997) finally reduces every mesh to at most 4000 vertices, so that all cases reach the cache at a comparable size.

Plausibility checks. Marching cubes reproduces whatever surface the labels imply, even when that surface is not a plausible ventricle. A label stack can be incomplete, noisy, or come from a long-axis rather than a short-axis acquisition, and the resulting mesh is then valid geometry but poor anatomy. Seven checks are therefore applied before a mesh enters the cache, and a case that fails any of them is discarded, because training on implausible geometry would weaken the shape prior the model is meant to learn.

Three of the checks ask whether the stack really covers the ventricle. The cross-sectional area of the endocardium must decrease from the mid-cavity towards the apex, because the ventricle narrows in that direction; when the apical end is the wider one, base and apex have most likely been swapped, or the stack stops before the true apex. The most basal slice must have an area consistent with the valve-plane region, since a very small basal contour means the acquisition ended too early and the basal cap would then close the mesh at an arbitrary height. The ratio between the apex-to-base length and the largest short-axis diameter must stay inside a fixed range, which removes globally distorted shapes.

The remaining four checks ask whether the segmentation is internally consistent. The slice centroids must not jump sideways from one slice to the next, because that pattern points to motion or an orientation error and produces twisted surfaces. Within each slice, the endocardial and epicardial rings must stay smooth and roughly convex, since jagged or self-intersecting rings usually come from local segmentation failures that smoothing cannot repair. At every level the endocardium must lie inside the epicardium with a positive gap between them, because the opposite ordering is an impossible target for a decoder that models the wall as an outward offset. The average gap between the two rings must also stay within plausible limits, which removes cases where labels have merged or leaked into neighbouring tissue.

Contour and query-point extraction. Accepted meshes are centred and normalised together, then cut by 10 equally spaced axial planes after a 5% margin at the apex and at the base. The largest intersection loop at each plane is resampled to 60 points for each available tissue. These are the contour points observed by the encoder. Separately, 2048 near-surface and volumetric query points are sampled and labelled by mesh occupancy for decoder supervision. The contour input and the volumetric target are therefore two representations of the same derived mesh, rather than independent annotations.

Figure 3.5 provides a visual summary of the conversion from clinical labels to mesh-ready supervision.

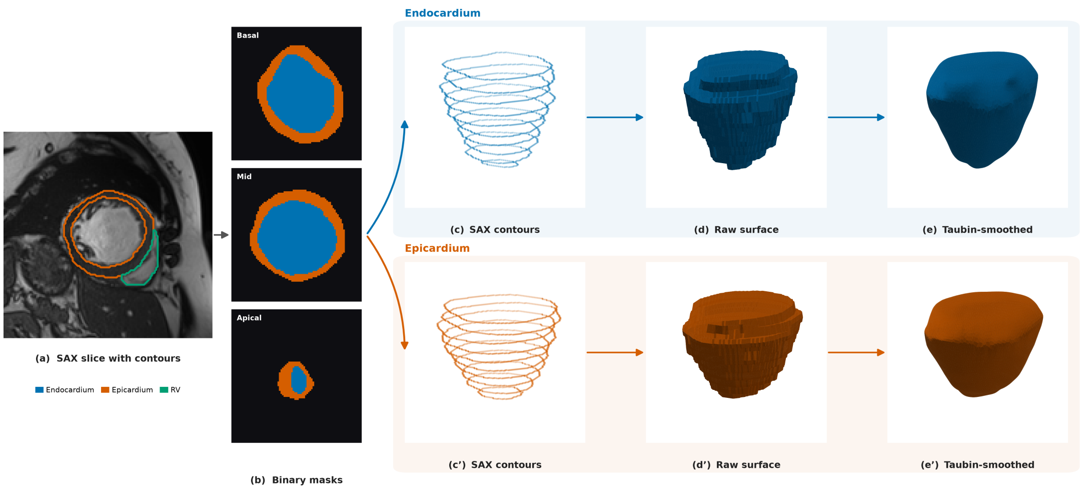

Figure 3.5. Real ED clinical-label-to-mesh pipeline (ACDC): from SAX labels and binary masks to endocardial and epicardial surface branches used for derived supervision.

Panel (a) shows the acquired SAX image with LV cavity, myocardium, and RV labels, and panel (b) shows the corresponding binary endocardial and epicardial masks after label harmonisation. The pipeline then branches into separate endocardial and epicardial paths: panels (c)–(e) and panels (c′)–(e′), respectively.

In both branches, contours are placed in physical three-dimensional positions, a voxel surface view makes discretisation artefacts explicit, and Taubin smoothing illustrates their reduction before later post-processing. The figure is intentionally a process overview; the full sequence also includes basal capping, cap fairing, decimation, the plausibility checks, and re-slicing.

3.2.5. Shared cache representation and contour augmentation

All cases are stored in a common cache representation so that the model receives the same interface regardless of whether a sample originates from the SSM or from a clinical segmentation.

A cache item combines the sparse contour observation with mesh-derived supervision and the metadata needed to keep both representations in a shared coordinate frame. The ED and ES caches contain 10 SAX levels, 60 points per available contour ring, and 2048 query points per case. This design deliberately separates what the model observes from what it is allowed to learn from: the encoder receives only contour coordinates, tissue identity, and cardiac phase, whereas the decoder is supervised by surface and volumetric samples derived from the corresponding mesh. Thus, at inference time, the same model can reconstruct geometry from sparse contour observations alone.

Table 3.1. Fields of one shared cache item and their role in training.

Field	Origin	Role
Contour points	Synthetic mesh slicing or clinical mask re-slicing	Sparse SAX observation for the encoder and contour-anchor loss
Tissue label	Contour extraction	Distinguishes endocardial from epicardial points during tissue-specific pooling
Phase label	Dataset metadata	Conditions the encoder on ED or ES geometry
Surface samples	Prepared endocardial and epicardial meshes	Surface-fitting and normal-alignment supervision
Query points and targets	Near-surface and volumetric mesh sampling	Dense SDF, occupancy, sign, and exterior supervision
Mesh metadata	Preprocessing pipeline	Coordinates, scale, faces, and orientation for evaluation and visualisation

Table 3.1 makes this separation explicit. The first three fields describe the clinical-style observation supplied to the encoder; the remaining fields are available only during training,

evaluation, or visualisation. Keeping these roles distinct prevents mesh-derived information from leaking into the encoder input while allowing dense supervision of the implicit field.

This common contract enables two-stage optimisation without dataset-specific changes to the architecture or loss. The model is first trained on synthetic ED samples, which establish a broad geometric prior, and is then fine-tuned on real ED and ES samples, which introduce patient-specific anatomy and phase-dependent shape variation. The loader changes only the source stream between stages; the fields consumed by the training loop remain unchanged.

Contour augmentation is applied only to the synthetic ED training observations, after the train/validation/test split. The target mesh, surface samples, query points, occupancy labels, tissue labels, and phase label remain fixed. This is therefore an observation-robustness intervention rather than the creation of a geometrically transformed anatomical target. Such label-preserving perturbations are commonly used to improve robustness when labelled data are limited (Chlap et al., 2021; Shorten & Khoshgoftaar, 2019).

For each augmented synthetic sample, the loader draws small independent in-plane translations per slice, point-wise in-plane jitter, global scale jitter, and contour-point dropout. Slice dropout and long-axis rotation are applied probabilistically while retaining at least three slice levels. The endocardial and epicardial contours within a slice receive the same translation, preserving their local relationship. As illustrated in Figure 3.6, these perturbations ask the encoder to recover one fixed mesh-derived target from plausible variations of its sparse SAX observation.

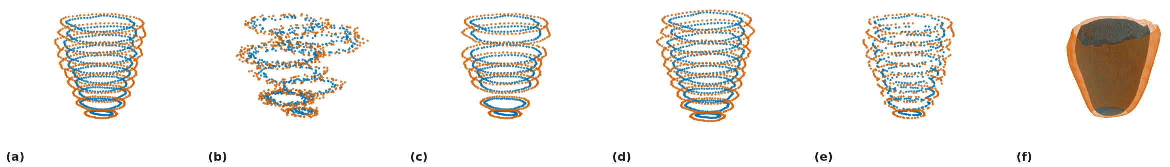

Figure 3.6. Examples of contour-input augmentation. Endocardial and epicardial points are shown in blue and orange, respectively.

In Figure 3.6, panel (a) shows the unaltered contour stack, while panels (b), (d), and (e) show slice translation, slice rotation, and contour-point dropout, respectively. Panel (f) shows the corresponding 3D mesh label. Thus, the sparse SAX observation changes across the augmented examples, whereas the derived mesh and all decoder-supervision targets remain fixed; the figure illustrates robustness of the encoder input rather than transformation of the anatomical ground-truth geometry.

3.2.6. Dataset composition and splits

This subsection reports how many samples each data stream finally contributes and how the real streams are divided for training and evaluation. The real data come from two public collections: the Automated Cardiac Diagnosis Challenge (ACDC) and the second Multi-Centre, Multi-Vendor and Multi-Disease Cardiac Segmentation challenge (M&Ms-2). Together they provide 460 patients (100 from ACDC and 360 from M&Ms-2). For each patient we reconstruct one ED mesh and one ES mesh, giving two real streams. Both collections are openly available for research at no cost; each requires a free registration and the acceptance of a data-use agreement, and both ask that the source publication be cited (Bernard et al., 2018; Campello et al., 2021). The synthetic stream is generated from the UK Digital Heart Project SSM: of its 100 modes, the first 32 already capture 99% of the shape variance, so we sample within that sub-space. Drawing shape coefficients under the Mahalanobis bound of Equation (3.2) and rejecting implausible shapes yields 1300 accepted meshes out of 2048 candidates (63%). The shape model is likewise open and free: it is distributed as a public repository by the UK Digital Heart Project,[1] needs no access application, and the authors ask only that the atlas papers be cited. None of the three sources used in this work therefore required a licence fee.

Not every real patient produces a usable mesh. A segmentation may be a long-axis view, cover too few slices, or fail an anatomical plausibility check, such as an endocardium that is not fully enclosed by the epicardium. After these quality gates, 444 of the 460 patients yield a valid ED mesh and 423 yield a valid ES mesh. The surviving meshes are then divided for training and evaluation. To prevent information from leaking between the partitions, the real streams are split by patient rather than by mesh, so that all meshes from a given patient stay within a single partition. Both streams use the same 70 %/15 %/15 % allocation for training, validation, and test, rounded to whole patients, and the synthetic ED meshes are used only to enrich the pre-training stage and are never placed in the validation or test sets.

1

https://github.com/UK-Digital-Heart-Project/Statistical-Shape-Model

Table 3.2. Final sample counts and patient-level partitioning of the three data streams. Real streams report patients that pass the quality gates; the synthetic stream reports meshes accepted out of the 2048 sampled shape vectors. Percentages in parentheses are the nominal allocation, rounded to whole patients.

Stream	Source	Phase	Count
Real data	ACDC and M&Ms-2	ED	444 patients
Real data	ACDC and M&Ms-2	ES	423 patients
Synthetic data	UK Digital Heart SSM	ED	1300 meshes

Table 3.2 records the two real source collections and the synthetic pre-training stream. The role of each stream in the two-stage training procedure is summarised in Appendix A.

3.3. Model Architecture

The model is built from two complementary parts. The encoder reads the sparse contour points and summarises the case into a compact latent representation, while a local contour-volume branch keeps the surrounding spatial structure in a coarse 3D feature field. The decoder then combines the global latent code, the local feature at each query point, and the point coordinates to predict the LV endocardial signed-distance field. Because the decoder can be evaluated anywhere in space, the reconstructed surface is not tied to a fixed grid or a fixed number of vertices, and a mesh is only generated once the field has been sampled densely. The output is then a continuous signed-distance representation of the chamber and its surrounding wall.

This design matches the structure of the problem. The contour input is a small, unordered set of points, so the encoder uses a point-cloud formulation that is invariant to the order of the points. The output is a smooth, continuous surface, so the decoder is an implicit function that can be evaluated at any resolution. The epicardium is defined as an outward offset from the endocardium, so the wall remains positive everywhere and the model learns a valid myocardial thickness field instead of two independent surfaces.

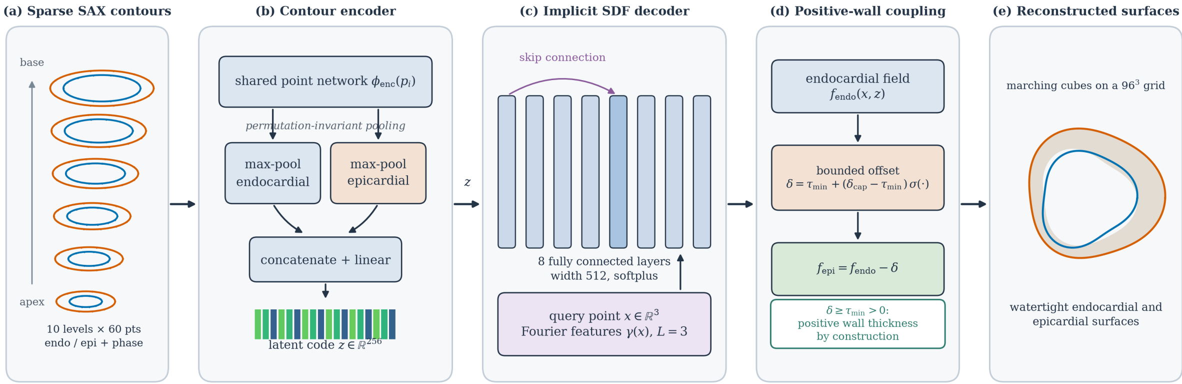

Figure 3.7. High-level view of the model. (a) Sparse endocardial and epicardial SAX contours, together with a tissue and a cardiac-phase label, form the observation. (b) A shared per-point network and tissue-specific max-pooling produce a global latent code z, while a parallel local contour-volume branch builds a 3D feature field. (c) A Fourier-encoded query point, the global code, and the interpolated local features are passed to an eight-layer implicit network with a middle skip connection. (d) The decoder emits the endocardial signed-distance field and a bounded positive offset δ, from which the epicardial field is obtained and the myocardial wall is constrained to remain positive. (e) Marching cubes on a 96^3 grid yields the two watertight endocardial and epicardial surfaces.

Figure 3.7 summarises the model as five stages. The sparse SAX contour observation of panel (a) is reduced by the encoder of panel (b) to a single 256-dimensional latent code using separate endocardial and epicardial max-pooling, while the same point features are also scattered into a local 3D feature volume. The global code captures the overall LV shape, and the local volume keeps the spatial context that is useful for estimating wall thickness. Both are passed to the implicit decoder of panel (c), which is evaluated at arbitrary three-dimensional points and therefore represents the geometry continuously rather than on a fixed mesh. Panel (d) shows the key geometric constraint: the model predicts the endocardial field together with a strictly positive bounded offset, so the epicardium is always separated from the endocardium by a valid wall thickness. Panel (e) shows the final output: the two watertight endocardial and epicardial surfaces extracted by marching cubes. The encoder is formalised in Section 3.3.1 and the decoder in Section 3.3.2.

3.3.1. Contour encoder

The encoder receives, for each contour point, its 3D coordinates together with a label indicating whether the point is endocardial or epicardial and a label indicating the cardiac phase. A small shared network processes every point independently and turns it into a feature vector. The features of the endocardial points and the features of the epicardial points are then combined separately by taking, for each feature, its maximum over the points. This makes the latent code invariant to the order of the contour points, which is important because contour points have no natural ordering across patients or slices. The two resulting descriptors are joined and projected to a single 256-dimensional latent vector z. At the same time, the same point features are scattered into a local 3D feature volume that keeps spatial neighbourhood information around the LV wall.

More formally, let $P = \{p_i\}_{i=1}^{N}$ be the contour points of one case, where each p_i carries coordinates, a tissue label, and a phase label. The shared point network computes

$$h_i = \phi_{\text{enc}}(p_i), \tag{3.5}$$

where ϕ_{enc} is the shared network. The endocardial and epicardial descriptors are then obtained by max-pooling within each tissue,

$$z_{\text{endo}} = \max_{i \in \mathscr{E}} h_i, \qquad z_{\text{epi}} = \max_{i \in \mathscr{P}} h_i, \tag{3.6}$$

where $\mathscr{E}$ and $\mathscr{P}$ are the endocardial and epicardial point sets. The global latent code is then

$$z = W_z[z_{\text{endo}}, z_{\text{epi}}] + b_z, \tag{3.7}$$

while the same scattered features are aggregated into a local 3D volume V and sampled at each query point by trilinear interpolation,

$$v(x) = \text{Interp}(V, x). \tag{3.8}$$

This local field adds spatial detail to the decoder while the global latent vector keeps the overall chamber shape compact and stable.

Pooling the two tissues separately keeps their information distinct. The endocardial points describe the cavity, while the epicardial points describe the outer boundary of the wall. If all points were pooled together, these roles would be mixed and the network would have a harder time separating the cavity from the wall. Keeping them separate gives the latent code clear access to both surfaces, while the local volume gives the decoder additional context at the myocardial boundary.

3.3.2. Implicit decoder and positive-wall coupling

Before a query point is given to the decoder, its coordinates are expanded with a Fourier positional encoding. This simply adds a few sine and cosine copies of the coordinates at different frequencies, which helps the network represent fine geometric detail instead of only smooth, blurry shapes (Tancik et al., 2020). Three frequency bands are used. For a query point $x \in \mathbb{R}^3$

the encoding is

$$\gamma(x) = [x, \sin(2^0 \pi x), \cos(2^0 \pi x), \ldots, \sin(2^{L-1} \pi x), \cos(2^{L-1} \pi x)], \tag{3.9}$$

with $L = 3$. The decoder itself is an eight-layer fully connected network of width 512 with a skip connection at the middle layer. It uses a smooth activation (softplus) so that the reconstructed surface is smooth and free of the sharp creases that a ReLU activation would introduce.

The decoder produces two outputs. The first is the signed distance to the endocardium, f_{endo}. The second is a positive offset δ, which represents the local wall thickness: how far the epicardium lies outside the endocardium at each query point. The decoder also receives the local contour feature $v(x)$ from the spatial volume, which is injected into the model so that it can use local wall context when predicting thickness. The offset is squeezed into a fixed positive range by a sigmoid,

$$\delta(x, z, v) = \tau_{\min} + (\delta_{\text{cap}} - \tau_{\min})\sigma\left(g_\delta(x, z, v)\right), \tag{3.10}$$

where g_δ is the raw decoder output, σ is the sigmoid, and $\tau_{\min}$ and δ_{cap} are the smallest and largest allowed wall thickness. The epicardial field is then

$$f_{\text{epi}}(x, z, v) = f_{\text{endo}}(x, z, v) - \delta(x, z, v). \tag{3.11}$$

Because Equation (3.10) always keeps $\delta \geq \tau_{\min} > 0$, the offset cannot be zero or negative. This prevents a collapsed wall in the field, but it does not show that the thickness is anatomically correct or that the extracted mesh has the correct topology after discretisation. In the final configuration $\tau_{\min} = 0.05$ and $\delta_{\text{cap}} = 0.45$ in normalised units, which corresponds to roughly 1.25–11.25 mm once mapped back to millimetres.

Table 3.3. Main architecture settings of the model.

Component	Configuration
Input features	3D coordinates, tissue label, and cardiac phase
Global encoder type	Shared network with endocardial and epicardial max-pooling
Local conditioning	Coarse 3D contour feature volume with trilinear interpolation
Latent dimension	256
Local volume size	16^3 with 32 channels
Positional encoding	Fourier features with $L = 3$ frequency bands
Decoder	8-layer network, width 512, skip connection at layer 4
Activation	Softplus (smooth)
Local injections	Local feature paths into the decoder trunk and wall head
Output heads	Endocardial signed distance and positive wall-thickness offset
Thickness range	$\tau_{\min} = 0.05$, $\delta_{\text{cap}} = 0.45$ in normalised units
Inference grid	96^3 query grid for marching cubes

3.4. Training and Inference

This section explains how the model is trained from the shared cache and how a surface mesh and a wall-thickness map are produced once training is finished. Training proceeds in two stages: the model is first pre-trained on the synthetic SSM cases and then fine-tuned on the real ED and ES cases. In both stages it must learn two things together: a valid distance field for the endocardium, and a sensible relationship between the endocardium and the epicardium.

3.4.1. Training data mix and augmentation

Training follows the two-stage strategy introduced in Section 3.2. In the first stage, the model is pre-trained only on the synthetic ED cases generated from the SSM. This large, controlled corpus teaches a strong prior for LV shape and tolerates aggressive augmentation. In the second stage, the pre-trained model is fine-tuned on the real cases reconstructed from ACDC and original M&Ms, covering both the ED and ES phases. Fine-tuning starts from the pre-trained weights and continues with the same architecture and loss, so the model keeps the geometric regularity learned from the SSM while adapting to real segmentation quality and to the systolic phase. Throughout both stages the phase label is provided as input, so a single model handles ED and ES, and augmentation is applied only to the contour input; the target meshes and the cached supervision are never changed.

A single phase-conditioned model is used instead of two separate models. The main reason is practical. When the experiments reported here were carried out, the public repository of the UK Digital Heart Project contained only the end-diastolic shape model, so a separate end-systolic model could not have benefited from the synthetic pre-training at all. An end-systolic shape model has since been added to that repository, but only after the training described in this chapter had been completed. Sharing one model lets the end-systolic cases reuse the shape prior learned from the synthetic end-diastolic corpus, and the phase label then tells the encoder which of the two configurations it is looking at. The cost of this choice is that the phase information is coarse. The model receives a binary label, not a continuous position in the cardiac cycle, so it can only represent two fixed states rather than the gradual change between them. Since the synthetic corpus is end-diastolic, the prior is also biased towards diastolic shape. Both effects make the end-systolic reconstruction the harder of the two cases, and this is visible in the results of Chapter 4.

The two stages are complementary. The synthetic ED cases give a large set of anatomically valid shapes but do not reproduce every artefact seen in real segmentations, which is why they are used for pre-training rather than as the final training data. Their role is to provide a geometric prior: they regularise the shape space, expose the model to a wide range of plausible LV geometries, and do so under controlled conditions. They do not provide realistic patient-specific wall thickness, because the epicardium in those samples is constructed by an offset rule. The real ED and ES cases then supply genuine patient anatomy from several datasets and let the model learn how the shape changes between the two phases. To avoid information leaking between the training, validation, and test sets, the real cases are split by patient before any augmentation is applied. Because the augmentation only perturbs the observed contours, each real target stays tied to the anatomy of its own patient.

In practice, this means the encoder is trained to be robust to imperfect, noisy inputs, such as small shifts, jitter, missing slices, or dropped points, while the decoder is always supervised against a fixed, consistent 3D geometry.

3.4.2. Loss function

The training objective contains five complementary terms. Surface and dense SDF supervision fit the geometry, Eikonal regularisation preserves a valid distance field, off-surface repulsion suppresses spurious zero crossings, and direct wall-thickness supervision constrains the myocardial wall. The full objective is

$$\mathscr{L}_{\text{total}} = \lambda_{\text{surf}}\mathscr{L}_{\text{surf}} + \lambda_{\text{SDF}}\mathscr{L}_{\text{SDF}} + \lambda_{\text{eik}}\mathscr{L}_{\text{eik}} + \lambda_{\text{off}}\mathscr{L}_{\text{off}} + \lambda_{\text{WT}}\mathscr{L}_{\text{WT}}. \tag{3.12}$$

The surface term places the two zero-level sets on their target meshes. The SDF term provides dense volumetric supervision at cached query points. The Eikonal term encourages the field gradient to have unit magnitude (Gropp et al., 2020), while off-surface repulsion keeps free-space samples away from either zero-level set. Finally, the wall-thickness term directly supervises the predicted offset δ at endocardial surface points. The Eikonal term is computed in full precision to keep its gradient calculation numerically stable.

Table 3.4. Purpose of the five training-loss terms.

Term	Purpose	Effect on reconstruction
$\mathscr{L}_{\text{surf}}$	Fits zero-level sets to surface samples	Places endocardial and epicardial surfaces on the target meshes
$\mathscr{L}_{\text{SDF}}$	Fits cached query-point distances	Provides dense volumetric supervision
$\mathscr{L}_{\text{eik}}$	Encourages $\|\nabla f\| = 1$	Regularises the decoder as a signed-distance field
$\mathscr{L}_{\text{off}}$	Repels off-surface points from zero	Reduces spurious zero crossings away from the myocardium
$\mathscr{L}_{\text{WT}}$	Fits the predicted wall offset to wall targets	Constrains myocardial wall thickness

3.4.3. Optimisation and hardware

Both the SSM pre-training stage and the real-data fine-tuning stage use the same optimisation setup: the AdamW optimiser at a learning rate of 2×10^{-5} and weight decay 5×10^{-4}, with the learning rate following a cosine schedule and early stopping ending each stage once the validation loss stops improving. Fine-tuning simply resumes from the pre-trained weights rather than starting from scratch. To fit the model in the available GPU memory, training uses mixed precision, gradient clipping at 1.0 for stability, and gradient accumulation to reach a larger effective batch size. Each stage runs for at most 400 epochs, though early stopping usually ends it sooner. Training was carried out on two GPUs in parallel. A light weight-clipping strategy was used instead of the more expensive spectral normalisation, since the decoder is evaluated at many points per step and GPU memory is the main constraint.

Table 3.5. Training configuration used for the model.

Setting	Value
Training strategy	SSM pre-training, then fine-tuning on real ED/ES
Training mode	Combined ED and ES with phase conditioning
Augmentation	Contour-input perturbations for synthetic and real cases
Optimiser	AdamW
Learning rate	2×10^{-5}
Weight decay	5×10^{-4}
Batch size	8, with gradient accumulation
Gradient clipping	1.0
Precision	Automatic mixed precision
Hardware	2 GPUs in parallel
Inference grid	96^3

3.4.4. Inference and analytic wall-thickness extraction

At inference time, the model turns the input contours into the latent vector z and then evaluates the decoder on a regular 96^3 grid of points. Marching cubes reads the zero-level sets of f_{endo} and f_{epi} from this grid and produces the raw endocardial and epicardial meshes (Lorensen & Cline, 1987). For evaluation, degenerate faces are removed, the largest connected component is retained, holes are filled, and a mesh is remade from a cleaned occupancy mask when needed. This post-processing makes the evaluation geometry usable for surface and thickness measurements; it is not a guarantee provided by the neural architecture. A wall-thickness value is available for free at every endocardial vertex, because the decoder already predicts the offset δ there; converting it to millimetres gives

$$t_{\text{model}}(x) = s\,\delta(x, z), \tag{3.13}$$

where s is the scale that maps normalised coordinates back to millimetres. This thickness is not measured after the fact by comparing two meshes; it is a direct output of the model. The selected measurement algorithms in Section 3.5 are therefore used as independent points of comparison on the same geometry, not as the only way to obtain thickness.

3.5. Wall-Thickness Evaluation Protocol

The wall-thickness evaluation is performed on the model-generated geometry. The segmentation volume is used to extract contour rings and to orient the AHA regional coordinate system, but the measured surfaces are the endocardial and epicardial meshes produced by the trained model. This distinction is important: the evaluation tests thickness estimation on the reconstructed implicit model, not directly on raw segmentation meshes. Myocardial wall thickness is itself a routine indicator of cardiac health, because a wall that is abnormally thin can signal infarction or scarring, while an abnormally thick wall can signal hypertrophy. The evaluation is therefore organised in three steps: first the algorithms that measure thickness on the geometry, then the resulting local three-dimensional thickness field, and finally its aggregation into the clinical AHA-17 regional summary.

3.5.1. Selected thickness methods

Four wall-thickness algorithms are evaluated on the reconstructed geometry. They are chosen to span the main methodological families used to measure thickness: volumetric distance fields, ray intersections, and partial-differential-equation (PDE) formulations. The four are the Laplace field method, the Yezzi–Prince Eulerian PDE method, the signed-distance cone-ray method, and the Euclidean Distance Transform (EDT) boundary sum. No method is assumed in advance to be the best one. All four are computed on the same geometry and compared under the same conditions, and the one that proves most consistent is then used for the regional analysis reported in Chapter 4. Each method returns local thickness values in millimetres at endocardial vertices or projected endocardial locations. The outputs are summarised with mean, median, standard deviation, fifth percentile, ninety-fifth percentile, finite-value fraction, and runtime, and are visualised as 3D colour maps and AHA-17 bullseye plots.

Table 3.6. Wall-thickness methods selected for evaluation on the model-generated LV geometry, with the Euclidean Distance Transform boundary sum kept only as a baseline.

Method	Category	Main measurement principle
Laplace field	PDE	Transmural field lines from a harmonic potential
Yezzi–Prince	PDE	Eulerian transport equations along the transmural field
SDF cone rays	Ray / SDF	Median of multiple cone rays around the surface normal
EDT boundary sum	Volumetric	Sum of endocardial and epicardial distance transforms

The EDT boundary sum computes, at each myocardial voxel x, the sum of the distances to the endocardial and epicardial surfaces,

$$t^{\mathrm{EDT}}(x) = D_{\mathrm{endo}}(x) + D_{\mathrm{epi}}(x), \tag{3.14}$$

where D_{endo} and D_{epi} are the Euclidean distance transforms of the two boundaries. This is exact for planar parallel walls but is limited by voxel resolution and slightly overestimates thickness in strongly curved regions.

The Laplace field method defines a smooth scalar field ψ over the myocardium by solving

$$\nabla^2\psi = 0, \qquad \psi|_{\mathrm{endo}} = 0, \qquad \psi|_{\mathrm{epi}} = 1. \tag{3.15}$$

The local wall thickness is then estimated from the gradient magnitude along the transmural field. This family of methods is attractive because harmonic field lines do not cross, but numerical errors can appear where the wall is thin or where the field gradient is small (Jones et al., 2000). The Yezzi–Prince approach addresses this by solving Eulerian transport equations along the same transmural field and combining the endocardial and epicardial travel times (Yezzi & Prince, 2003). In simplified form,

$$\nabla\psi \cdot \nabla u = -1, \qquad t = u + v, \tag{3.16}$$

where u and v are the propagation distances from opposite boundaries.

The cone-ray method uses the endocardial surface normal but reduces sensitivity to a single noisy ray. Inspired by the shape diameter function (Shapira et al., 2008), for each endocardial point several rays are cast inside a cone around the normal direction, and the median valid hit distance is used:

$$t_i^{\text{cone}} = \text{median}_{k=1}^{K} d_{ik}, \qquad K = 7. \tag{3.17}$$

In the notebook implementation the cone half-angle is 30°. This makes the estimate more robust than a single normal ray while still preserving an approximately transmural interpretation.

Figure 3.8 contrasts how the four estimators measure thickness on the same wall cross-section. The EDT method sums the distances from an interior point to both boundaries; the Laplace method follows the transmural streamlines of a harmonic potential whose nested iso-surfaces are shown as dashed lines; the Yezzi–Prince method integrates two travel times u and v along that same field and adds them; and the cone-ray method casts a fan of rays around the endocardial normal and keeps the median hit distance.

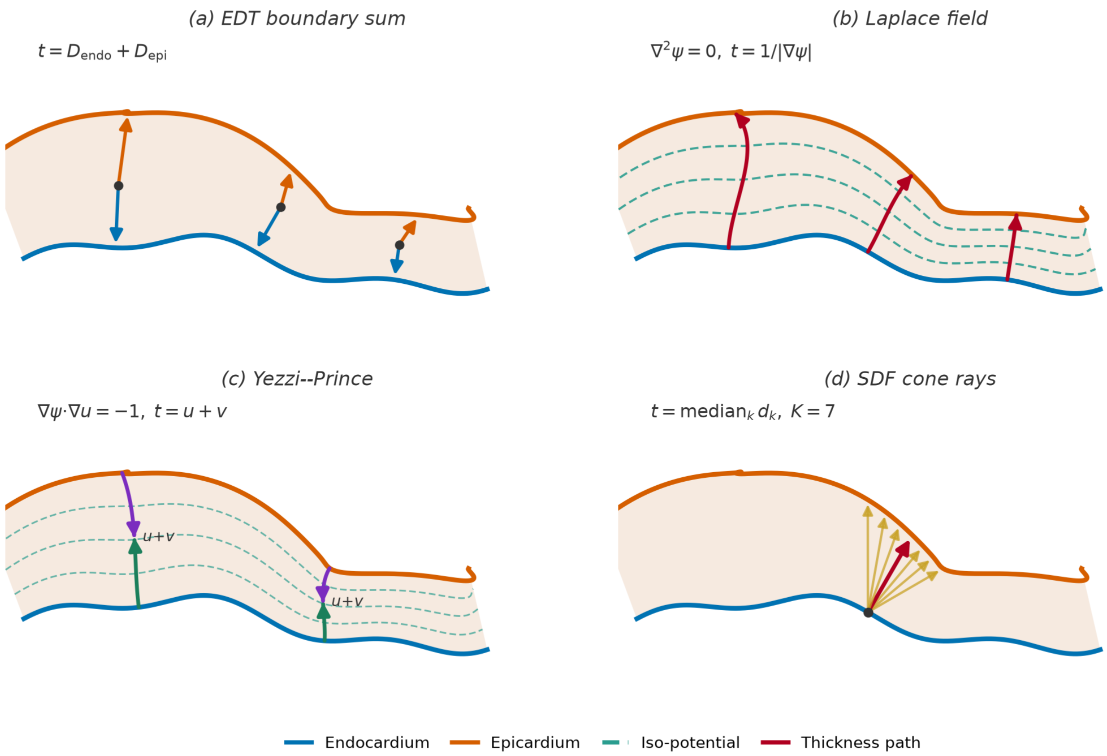

Figure 3.8. Conceptual comparison of the four wall-thickness estimators on a single myocardial cross-section, with the endocardium in blue and the epicardium in orange. (a) The EDT boundary sum adds the distances from an interior point to each boundary. (b) The Laplace field method follows the curved transmural streamlines of a harmonic potential, which cross the iso-potential surfaces (dashed lines) at right angles, and estimates thickness from the gradient magnitude. (c) The Yezzi–Prince method propagates two travel times u and v from opposite boundaries along that same curved field and sums them. (d) The SDF cone-ray method casts a fan of rays within a cone around the endocardial normal and takes the median intersection distance (highlighted), making it robust to any single ill-oriented ray.

3.5.2. Local three-dimensional wall thickness

Each of the selected methods produces, before any regional summary, a continuous thickness field defined over the whole reconstructed geometry. Because the model decoder can be evaluated at any point in space, the endocardial and epicardial surfaces are watertight and densely sampled, and a thickness value is attached to each endocardial vertex. This produces a local three-dimensional wall-thickness map rather than a handful of per-slice numbers, as shown in Figure 3.9, where the LV surface is coloured by local thickness from the thin apical region to the thicker basal wall.

This local map is the primary output of the wall-thickness stage. Presenting the measurement first as a full surface field, and only afterwards as regional segment averages, makes the spatial distribution of thickness explicit: localised thinning or thickening remains visible on the surface even when it would be averaged away inside a single AHA segment. The colour map uses a fixed thickness range across cases so that maps from different patients and cardiac phases can be compared directly.

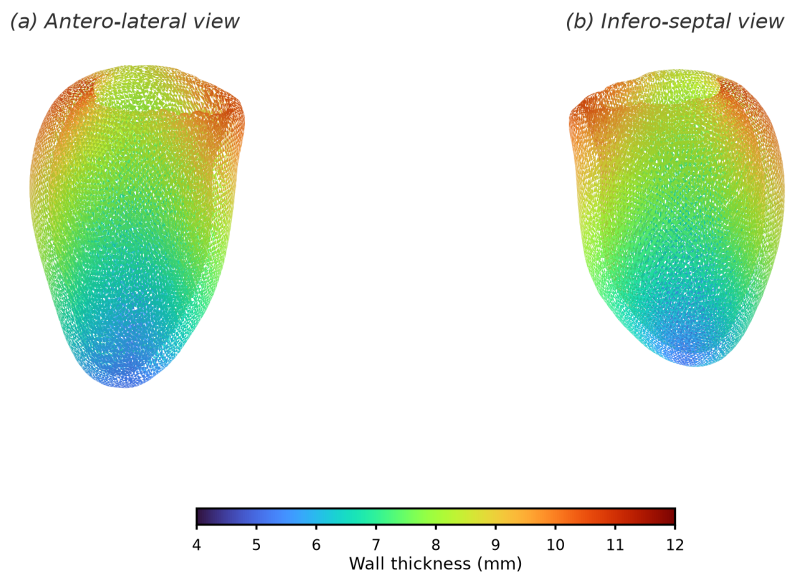

Figure 3.9. Local three-dimensional wall-thickness map on the reconstructed LV endocardial surface, shown from two opposing viewpoints. Each vertex is coloured by its local wall thickness on a fixed millimetre scale, so that the continuous spatial variation from the thin apex to the thicker base is visible directly on the geometry rather than only as regional averages.

3.5.3. Regional AHA-17 mapping

The dense surface field is finally aggregated into the AHA 17-segment model, the accepted clinical language for reporting regional myocardial measurements (Cerqueira et al., 2002). Conventionally the wall thickness in each segment is read semi-manually from a few SAX slices, which is observer-dependent, coarsely sampled, and sensitive to the choice of slice and measurement point. Here the same segment values are produced objectively from the local field of Section 3.5.2, by cutting the reconstructed surface into the AHA segments and averaging within each one.

The cut is defined in the LV's own anatomical frame, which first requires identifying the long-axis direction, distinguishing base from apex, and fixing a septal reference direction; the wall-thickness notebook includes an anatomical QA step that visualises this base/apex orientation and septal alignment before the segments are formed. The surface is then partitioned in two stages, as illustrated in Figure 3.10. First, along the long axis, the endocardial vertices are split by their normalised apex–base position into a basal ring, a mid-cavity ring, an apical ring, and a small apical cap. Second, within each ring the vertices are divided angularly around the long-axis centreline: the basal and mid rings are cut into six sectors each and the apical ring into four, while the apical cap forms the single seventeenth segment. Numbering the sectors from the anterior–septal reference and unrolling the rings onto concentric circles yields the familiar bullseye, in which the basal ring is the outer annulus, the mid ring the middle annulus, the apical ring the inner annulus, and the apical cap the central disc.

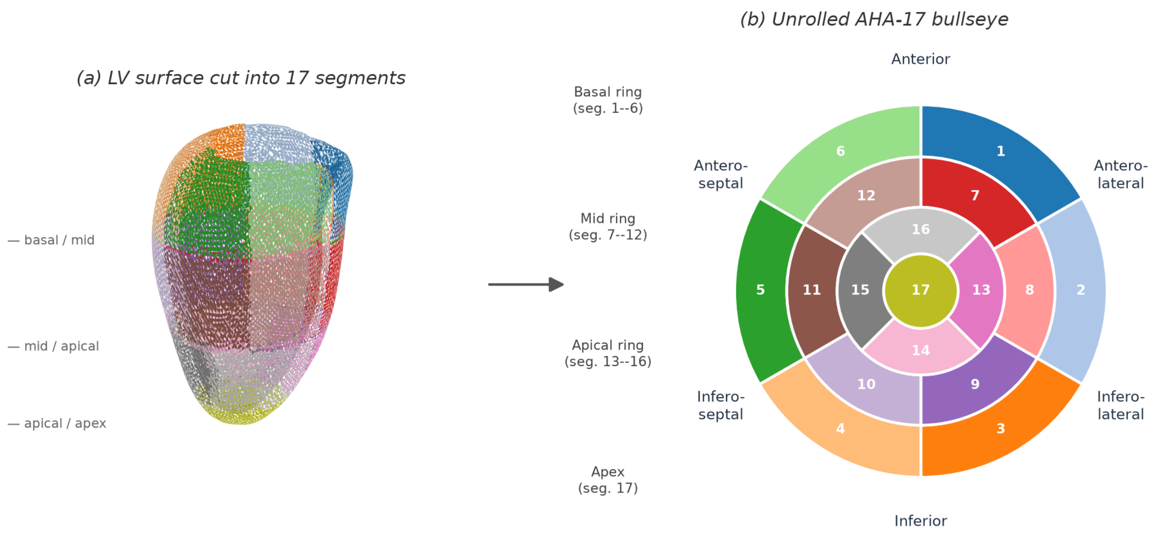

Figure 3.10. Cutting the reconstructed LV surface into the AHA-17 segments. Panel (a) shows the endocardial surface partitioned along the long axis into basal, mid, apical, and apical-cap levels and angularly into sectors, with each of the 17 regions drawn in a distinct colour. Panel (b) unrolls the same colour-coded regions onto the standard AHA-17 bullseye, where the basal ring (segments 1–6) forms the outer annulus, the mid ring (7–12) the middle annulus, the apical ring (13–16) the inner annulus, and the apical cap (17) the central disc; the anatomical wall regions are labelled around the disc.

The AHA aggregation does not replace the per-vertex thickness maps; it provides a clinically familiar regional summary of the same local measurements. For each method, the thickness values of the vertices falling in a segment are combined using robust summaries such as the mean and percentiles, so that local outliers do not dominate the regional interpretation.

This evaluation should be interpreted as an algorithmic comparison and reproducibility study rather than as clinical validation. Unless expert manual wall-thickness measurements are added, the thesis can conclude that the model provides an objective and locally resolved computational framework, but it should not claim replacement of clinical measurement protocols.

CHAPTER 4

Results and Evaluation

This chapter presents the experimental evaluation of the approach described in Chapter 3.

4.1. Experimental Setup

The evaluation uses the final model, obtained by pre-training on the synthetic SSM meshes and then fine-tuning on the real end-diastolic (ED) and end-systolic (ES) data, as described in Chapter 3. The datasets and patient-level splits are reported in Section 3.2.6. The results below are computed on a cohort of 30 ACDC patients, comprising 20 normal (NOR) and 10 hypertrophic-cardiomyopathy (HCM) cases: each is reconstructed at end-diastole for the geometric and wall-thickness comparisons, and at both phases for the regional analysis. The HCM group is included so that the model can be examined on abnormal as well as on normal left ventricles. Reconstruction accuracy is measured against a segmentation-derived baseline obtained by marching-cubes extraction on the input segmentations, which is the only reference available for these scans. Wall thickness is then measured on the reconstructed geometry alone. Two questions guide the evaluation: how faithfully the model reconstructs the LV surfaces from sparse contours, and how the four thickness algorithms behave on the surfaces it produces.

4.1.1. Metrics

Reconstruction quality is measured with five quantities, following the metric taxonomy of Taha and Hanbury (2015) and the conventions used for cardiac segmentation (Bernard et al., 2018) and for implicit-surface learning (Mescheder et al., 2019). Each one answers a different question: how large the error is at a typical point, how large it is in the worst region, how much of the enclosed volume is recovered, whether the reconstruction is systematically too big or too small, and whether the surface faces the right way.

The *Chamfer distance* is the average distance between the two surfaces, taken in both directions. It is reported for the endocardium and the epicardium in millimetres, and smaller is better. An average hides isolated failures, so the *95th-percentile Hausdorff distance* (HD95) is reported next to it. HD95 is the value below which 95% of the point-to-surface distances fall, so it describes the worst part of the surface without being set by a single outlying vertex.

The *Dice similarity coefficient* (DSC) measures the solid rather than the surface. After voxelisation it is twice the shared volume divided by the sum of the two volumes. It is computed for the endocardial cavity and for the myocardium, lies between zero and one, and larger is better. A shape can score well on overlap and still be uniformly too small, so the *volume ratio* is reported as well. It is the reconstructed volume divided by the reference volume, and a value of one means there is no systematic error in size. It is the only one of the five metrics that detects such a bias.

The *normal consistency* is the average absolute cosine similarity between the surface normals of corresponding points. It measures orientation rather than position, and therefore separates a smooth surface that follows the reference from a rough one that keeps crossing it. The *watertight rate* is the fraction of reconstructions that close into a surface with no holes. It is stated in the text as a precondition rather than as a score, because a mesh that is not closed has no volume for the DSC and the volume ratio to use.

Three metrics that often appear alongside these are left out. The average symmetric surface distance is almost identical to the Chamfer distance here, differing by at most 0.04 mm. The intersection-over-union is a direct transformation of the DSC. The F-score at a fixed tolerance repeats the surface-distance result at a single threshold. Reporting them would suggest more independent evidence than there is.

Wall thickness is summarised by its mean, its standard deviation, and its 5th and 95th percentiles across the myocardial surface, all in millimetres. The percentiles describe the thin apical and thick basal extremes without being distorted by isolated outliers. When the two cardiac phases are compared, the absolute thickening $\Delta_{\mathrm{ES-ED}}$ and the relative thickening as a percentage of the end-diastolic value are also reported. These are the standard descriptors of regional contractile function.

The four wall-thickness methods are compared with one another on the reconstructed geometry, using the agreement framework of Bland and Altman (1986) instead of a comparison of mean values. The reported quantities are the *Pearson correlation coefficient* r, the *mean absolute error* (MAE) and *root-mean-square error* (RMSE) in millimetres, and the Bland–Altman *bias* with its *95% limits of agreement* (bias $\pm 1.96\,\sigma$). Absolute consistency is summarised by the two-way *intraclass correlation coefficient* (ICC), read against the thresholds of Koo and Li (2016). These are reported together because correlation alone does not show agreement: two methods can rise and fall together and still differ by a constant amount, which only the bias reveals.

4.1.2. Baselines

Because no paired ground-truth 3D mesh exists for the clinical scans (see Section 3.2.1), the reconstruction is compared against meshes derived from the segmentations. Wall thickness is then measured on the reconstructed geometry with the four algorithms selected in Section 3.5.1: the Laplace field method, the Yezzi–Prince method, the SDF cone-ray method, and the Euclidean Distance Transform (EDT) boundary sum. The thickness computed directly on the input label mask is reported as an anchor for the overall magnitude, as the boundary sum of the two distance transforms over every myocardial voxel.

That anchor is not a ground truth, because the input is sparse. Across the cohort only 8.6 slices on average cross the myocardium, from 6 to 14 depending on the patient. The slices are 10 mm apart in 29 of the 30 cases and 5 mm apart in the remaining one, while the in-plane pixel size is between 1.37 and 1.88 mm. Resampling such a stack to a 1.0 mm isotropic grid interpolates about nine voxels in ten along the long axis, so the anchor carries almost no measured information in that direction. Absolute millimetre values should therefore be read with care throughout this chapter.

4.2. Training

The final model is produced in two stages. A first stage pre-trains the signed-distance decoder for roughly 200 epochs on the synthetic SSM meshes. This stage provides a geometric prior, and its checkpoint initializes the second stage. The reported model is then fine-tuned on the real end-diastolic and end-systolic data described in Chapter 3. The available training history records the fine-tuning stage; no numerical claim is made about the pre-training loss because its per-epoch log was not retained.

Fine-tuning runs for 776 epochs with the AdamW optimiser (learning rate 2×10^{-5}, weight decay 5×10^{-4}) under a cosine-annealing schedule with warm restarts, mixed-precision arithmetic, and an effective batch size of 16. The composite objective is the five-term loss of Equation (3.12), combining surface, dense SDF, Eikonal, off-surface, and wall-thickness supervision. Figure 4.1 plots the total loss and its principal components together with two optimisation diagnostics. The total training loss decreases from 10.07 at the first fine-tuning epoch to 0.91, and the validation loss settles at 0.70, with the best checkpoint selected at epoch 695 before mild over-fitting appears. The surface and Eikonal terms fall by more than an order of magnitude, confirming that the reconstructed surfaces converge onto the observed contours while the field approaches a true signed-distance function; consistently, the mean gradient norm $\|\nabla f\|$ stabilises at 0.996, essentially the unit value the Eikonal term enforces.

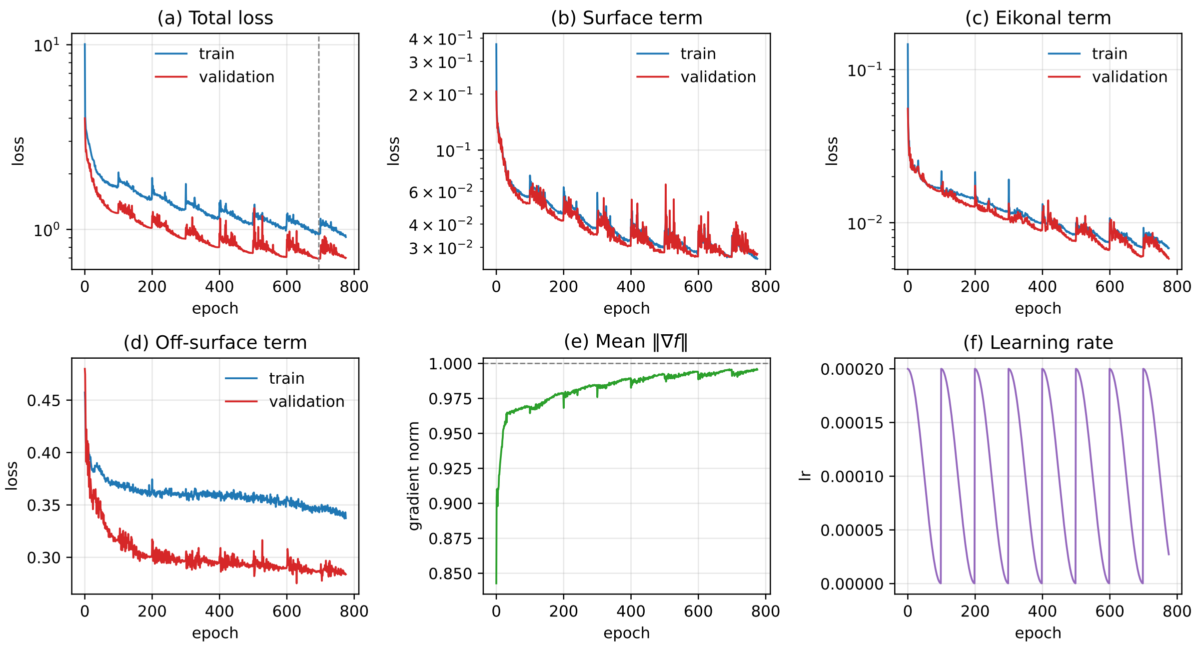

Figure 4.1. Fine-tuning history. Panels (a)–(d) show the total loss and its surface, Eikonal, and off-surface components for the training and validation splits; the dashed vertical line marks the best epoch. Panels (e)–(f) show the mean field gradient norm, whose target is one, and the cosine-annealing learning-rate schedule. The synthetic pre-training stage is not shown because its optimisation history was not retained.

4.3. Results

This section reports first the geometric quality of the reconstruction, then the wall-thickness measurements obtained from the reconstructed geometry, and finally the regional analysis over the AHA 17-segment model.

4.3.1. Reconstruction Quality

Table 4.1 summarises reconstruction accuracy over the evaluation cohort. Each case is reconstructed at end-diastole from its SAX contours and compared with the reference geometry. The five metrics are read in turn, because each one answers a question the others cannot.

The Chamfer distance gives the error at a typical point. It is 1.22 ± 0.21 mm on the endocardium and 1.05 ± 0.18 mm on the epicardium, so an average point of the reconstructed surface lies about one millimetre from the reference. This is finer than the in-plane pixel size of the source images, which is between 1.37 and 1.88 mm.

The HD95 gives the error in the worst region. It is 3.30 ± 0.91 mm on the endocardium and 3.47 ± 1.41 mm on the epicardium, about three times the corresponding mean. A worst case three times the average, rather than ten times, means the remaining error sits in a limited part of each surface instead of being spread over all of it. The larger standard deviation on the epicardium shows that this region varies more from patient to patient on the outer surface.

The Dice score gives the share of the enclosed volume that is recovered. It is 0.93 ± 0.02 for the cavity and 0.85 ± 0.02 for the myocardium. The myocardial value is lower because the myocardium is a thin shell. A boundary shift of one millimetre removes a much larger fraction of a wall a few millimetres thick than of a compact cavity, so the two numbers are not directly comparable and the gap between them is expected.

The volume ratio shows whether the reconstruction is systematically the wrong size. It is 0.92 ± 0.03 for the cavity and 1.00 ± 0.03 for the epicardial solid. The epicardium is therefore the right size on average, while the cavity is consistently about 8% too small. The Chamfer distance cannot show this, because it would report the same sub-millimetre error for a uniform inward shift as for random noise of the same size.

The normal consistency shows whether the surface faces the right way. It is 0.87 ± 0.05. A value close to one means the reconstructed normals follow the local orientation of the reference, so the small distances above come from a smooth surface lying along the reference rather than from a rough one crossing it. Every endocardial and epicardial reconstruction in the cohort also closes into a manifold, a watertight rate of 100% on both surfaces after the repair stage. This is what makes the enclosed volumes well defined, and therefore what makes the volume ratios meaningful.

Splitting the cohort by group shows that the reconstruction degrades only mildly on the abnormal ventricles. The endocardial Chamfer distance is 1.12 mm on the normal cases and 1.43 mm on the hypertrophic ones, the endocardial HD95 rises from 2.89 to 4.12 mm, and the endocardial Dice falls from 0.94 to 0.91. The myocardial Dice is slightly higher on the HCM group (0.87 against 0.84), because a thicker wall is easier to overlap. The clearest group effect is in the normal consistency, which falls from 0.89 to 0.83, consistent with the more irregular wall geometry of the hypertrophic cases.

Figure 4.2 shows the reconstructed endocardial and epicardial surfaces for one normal case at both cardiac phases, beside the geometry derived from the same segmentation. The end-systolic cavity is visibly smaller than the end-diastolic one, and the surrounding wall correspondingly thicker, which is the expected effect of contraction and confirms that the phase-conditioning input steers the decoder towards the correct geometry. Both reconstructed surfaces are zero-level sets of the same continuous field rather than stacks of per-slice contours, and the comparison with the segmentation-derived surfaces shows that the reconstruction recovers the same overall shape while smoothing the voxel staircase of the label mask.

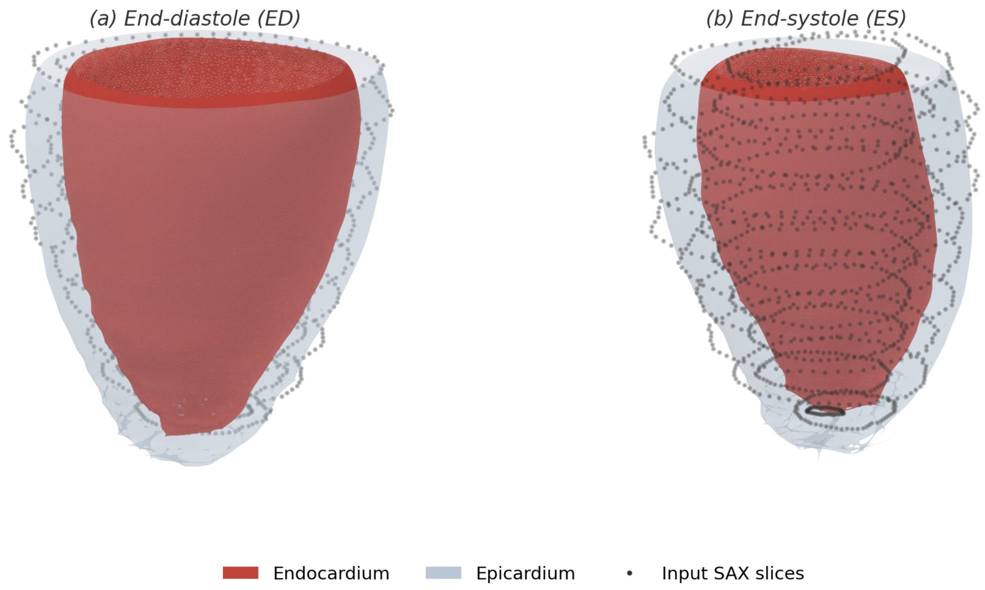

Figure 4.2. Reconstructed and segmentation-derived left-ventricular surfaces at both cardiac phases, for one normal case of the evaluation cohort. Panels (a) and (c) show the reconstruction, panels (b) and (d) the geometry derived from the same segmentation. The opaque red surface is the endocardium and the translucent grey shell is the epicardium; all eight surfaces are watertight, and the four panels share one scale and viewpoint, with the base upwards. The end-systolic cavity is smaller and the wall thicker, consistent with contraction. The fine circumferential banding on the reconstruction marks the through-plane spacing of the input SAX slices, between which the field is interpolated; the segmentation-derived surfaces instead show the voxel staircase of the label mask.

Table 4.1. Reconstruction quality of the final model over the 30-patient evaluation cohort at end-diastole, reported as mean $\pm$ standard deviation across patients. The five metrics cover, in order, the typical surface error, the worst-case surface error, the volumetric overlap, the systematic volume bias and the agreement in local orientation. Distances are in millimetres; the Dice similarity coefficient, the volume ratio and the normal consistency are dimensionless. Every reconstruction in the cohort is watertight on both surfaces.

Metric	Endocardium	Epicardium
Chamfer distance (mm)	1.22 ± 0.21	1.05 ± 0.18
HD95 (mm)	3.30 ± 0.91	3.47 ± 1.41
Volume ratio	0.92 ± 0.03	1.00 ± 0.03
	Cavity	Myocardium
Dice similarity coefficient	0.93 ± 0.02	0.85 ± 0.02
Normal consistency	0.87 ± 0.05	

4.3.2. Wall-Thickness Measurement

The four selected methods are applied to the reconstructed geometry of every patient at end-diastole. Table 4.2 reports, for each method, the cohort average of the per-patient mean, standard deviation, and 5th and 95th percentiles, and Figure 4.3 shows the same values graphically.

The three field-based methods agree on the magnitude of the wall. The Laplace field reads 9.18 mm, the EDT boundary sum 9.55 mm and the Yezzi–Prince method 10.01 mm, all within about one millimetre of the 8.96 mm anchor measured on the input label mask. The SDF cone-ray method stands apart at 12.47 mm, some 3.5 mm above the anchor.

Dispersion separates the methods the same way. The per-patient standard deviation is 1.64 mm for Yezzi–Prince, 1.66 mm for the EDT boundary sum and 2.14 mm for the Laplace field, but 4.87 mm for the cone-ray estimator. Its 5th-to-95th-percentile span, 8.01 to 20.68 mm, is about twice as wide as that of any other method. The ray-based estimator is therefore both the highest and the noisiest of the four.

Table 4.2. Wall-thickness statistics for the four selected methods on the reconstructed geometry, at end-diastole over the 30-patient evaluation cohort. Each entry is the cohort average of the corresponding per-patient statistic, in millimetres, before any calibration. For reference, the thickness measured directly on the input label mask averages 8.96 mm.

Method	Mean	Std.	p_5	p_{95}
Laplace field	9.18	2.14	4.75	12.20
Yezzi–Prince	10.01	1.64	7.41	12.74
SDF cone rays	12.47	4.87	8.01	20.68
EDT boundary sum	9.55	1.66	6.62	12.17

Table 4.3 compares the three other methods with the Laplace field point by point over the myocardial surface of all 30 patients. The two remaining field-based methods track it closely. The EDT boundary sum differs from it by 0.81 mm on average with a bias of +0.37 mm, and Yezzi–Prince by 0.89 mm with a bias of +0.83 mm; both reach ICC = 0.77. The cone-ray method differs by 3.38 mm on average, with a bias of +3.31 mm, limits of agreement of $\pm$12.74 mm and ICC = 0.28. Read against the thresholds of Koo and Li (2016), the first two show moderate to good agreement with the Laplace field and the third poor agreement. The correlations make the same point in a different way: all three exceed $r = 0.44$, but r alone would not have revealed a bias of more than three millimetres.

The Laplace field is the method used for the regional analysis of Section 4.3.3. The reason is methodological rather than numerical. It is the established reference for transmural thickness in cardiac imaging (Jones et al., 2000), and it measures thickness along the streamlines of a harmonic field spanning the two surfaces, so the path stays inside the wall wherever the wall curves. A straight ray does not: it leaves the myocardium early or late in curved regions, which is what the cone-ray numbers above show. The results here are consistent with that choice, since the Laplace field sits with the two other field-based methods rather than with the outlier. No claim is made that it is closer to true anatomy, because no independent measurement of the anatomy exists for these patients.

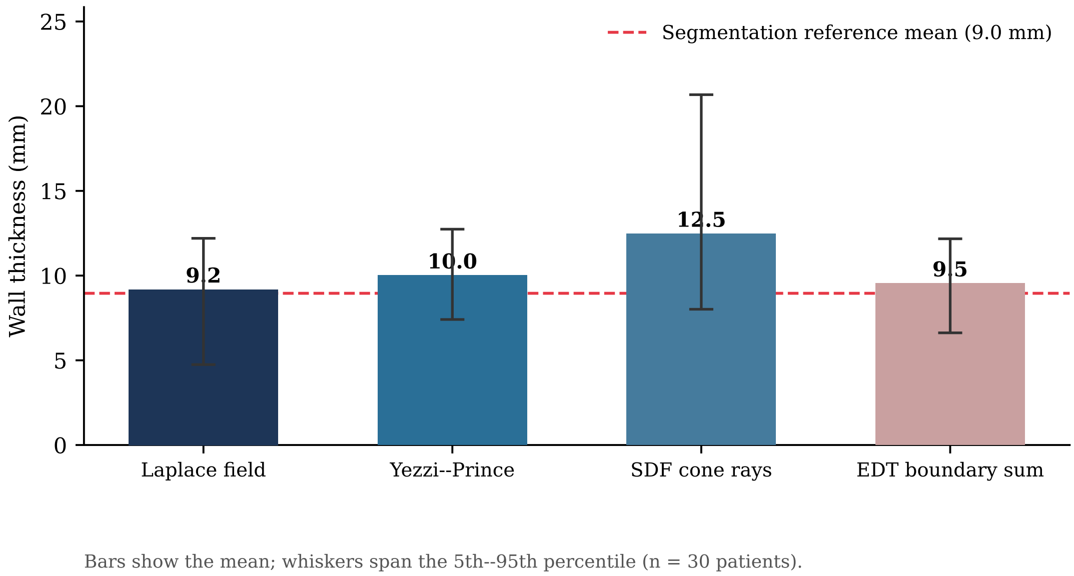

Figure 4.3. Mean wall thickness of the four selected methods on the reconstructed LV geometry, with whiskers spanning the 5th to 95th percentile. The dashed line marks the anchor value measured directly on the input label mask.

Because the decoder returns a continuous field rather than a set of per-slice values, the analytic thickness can be mapped back onto the reconstructed surface itself. Figure 4.4 shows this map for the representative case at both cardiac phases, in the same form as the methodology illustration of Figure 3.9 but computed on the model output instead of on the statistical-shape-model mean mesh. The decoder offset is defined only up to the normalisation applied to the input contours, so its absolute level is calibrated per phase to the mean of the distance-transform reference measured on the input segmentation, exactly as the cohort pipeline does; the calibration factor is close to two for both phases, and what the map contributes is therefore the spatial distribution rather than the overall magnitude. That distribution is anatomically plausible: the wall thins towards the valve plane and towards the apex and is thickest around the mid-cavity level, and the end-systolic surface is uniformly warmer than the end-diastolic one, which is the contraction-driven thickening that Table 4.4 summarises regionally. The residual circumferential banding visible on the surfaces marks the through-plane spacing of the input SAX slices, where the field is interpolated between observations.

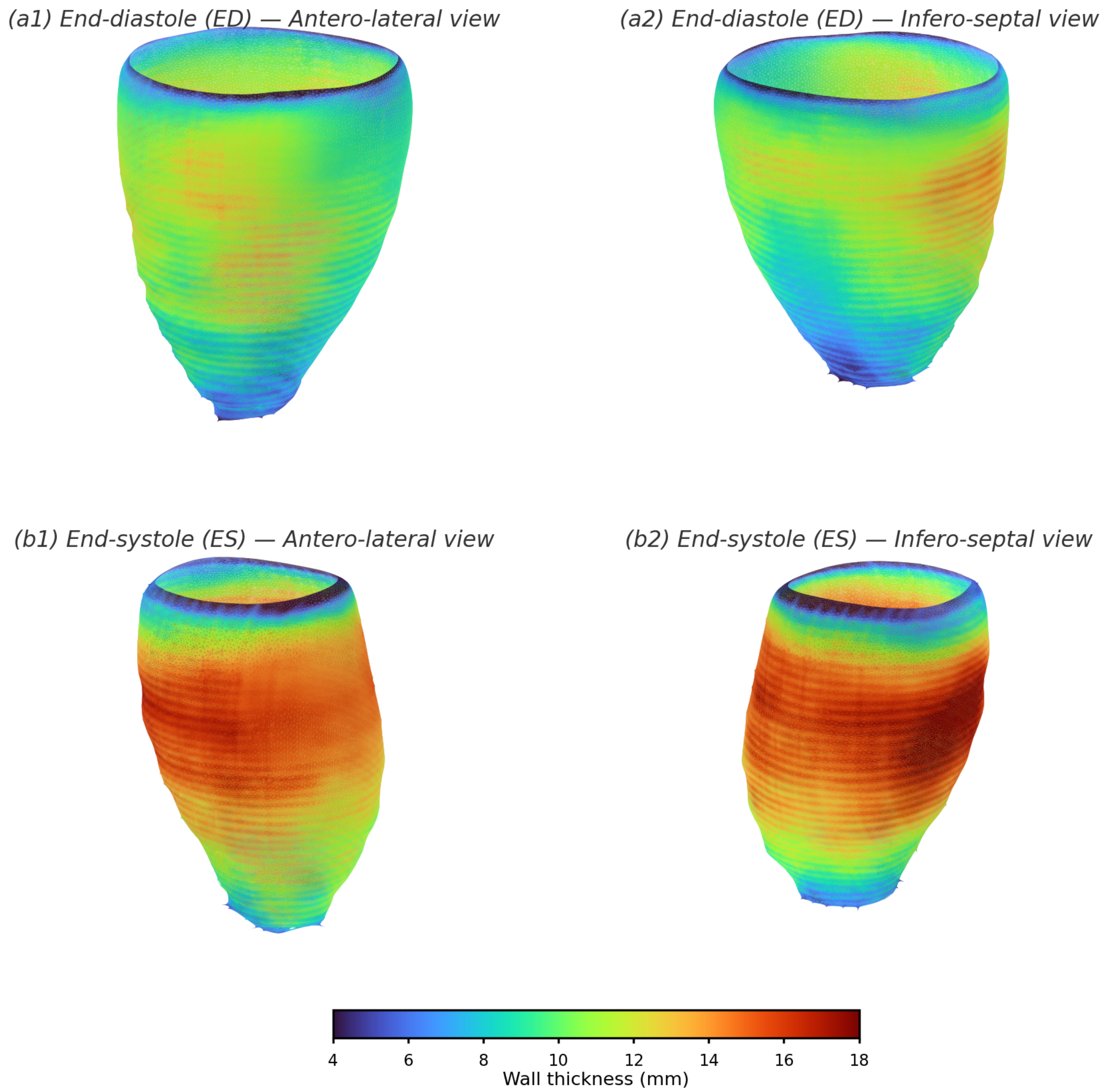

Figure 4.4. Analytic wall thickness mapped onto the reconstructed endocardial surface of one test case from the ACDC cohort, at end-diastole (top) and end-systole (bottom), each seen from two opposing viewpoints. Every vertex is coloured by the decoder wall offset evaluated at that point, calibrated per phase to the distance-transform reference of the input segmentation, on a scale shared by all four panels. The surface is trimmed at the most basal input contour, so the base ends in an open rim and only the observed part of the wall is shown; above that plane the field is pure extrapolation.

Table 4.3. Agreement of each wall-thickness method with the Laplace field, over the myocardial surface points of the 30 patients at end-diastole on the reconstructed geometry. MAE, RMSE, bias and the 95% limits of agreement (LoA) are in millimetres; the Pearson correlation r and the intraclass correlation coefficient (ICC) are dimensionless. The Laplace field is the common comparator, not a ground truth.

Method	r	MAE	RMSE	Bias (95% LoA)	ICC
Yezzi–Prince	0.80	0.89	2.14	+0.83 (±3.86)	0.77
SDF cone rays	0.44	3.38	7.29	+3.31 (±12.74)	0.28
EDT boundary sum	0.78	0.81	2.07	+0.37 (±3.99)	0.77

4.3.3. Regional Analysis over the AHA 17-Segment Model

To localise the measurements anatomically, the reconstructed surface is partitioned into the seventeen segments of the American Heart Association (AHA) model, assigning each myocardial vertex to a basal, mid-cavity, apical, or apical-cap segment from its normalised long-axis height and its circumferential angle relative to the right-ventricular insertion. Figure 4.5 shows the partition applied to the reconstructed geometry of the representative case together with the resulting bullseye map of mean wall thickness, and Table 4.4 reports, for every segment, the mean end-diastolic and end-systolic thickness on the reconstructed model, separately for the normal and the hypertrophic patients, together with the relative wall thickening between the two phases. The values are measured with the Laplace field, for the reasons given in Section 4.3.2. Splitting the table by group is what makes it possible to see whether the model reacts to an abnormal ventricle or simply reproduces an average one.

The regional pattern is anatomically consistent in both groups. In the normal cases the end-diastolic thickness varies little around the base and the mid-cavity, from 7.92 to 8.81 mm, and tapers towards the apex, where the apical cap at 6.80 mm is the thinnest region. Every segment thickens from end-diastole to end-systole, and the thickening is graded along the long axis: the apical segments thicken by 52% to 64%, the mid-cavity segments by 51% to 64% and the basal segments by 37% to 61%. Over the normal group the mean thickness rises from 7.99 mm to 12.21 mm, a thickening of 52.9%.

The hypertrophic group is thicker everywhere, which is the effect the comparison was designed to expose. Its cohort-average end-diastolic thickness is 11.66 mm against 7.99 mm in the normal group, a difference of 3.67 mm, and every one of the seventeen segments is thicker than its normal counterpart. The difference is largest in the septal and anteroseptal mid-cavity segments, where the wall reaches 13.99 and 13.43 mm against 8.55 and 8.57 mm in the normal group, which is the septal-dominant pattern typical of hypertrophic cardiomyopathy. The model therefore separates the two groups rather than regressing every case towards a common mean shape.

Systolic thickening behaves differently from absolute thickness. The hypertrophic group thickens by 47.8% against 52.9% in the normal group, so the two groups are much closer in relative terms than in absolute ones. This is consistent with hypertrophic myocardium being thicker at rest without contracting proportionally more. Taking the thickest segment of each patient as a simple severity index, the normal cases reach 9.07 mm on average and none exceeds the 15 mm clinical threshold, whereas the hypertrophic cases reach 14.44 mm and three of the ten exceed it. The direction of the effect is therefore correct, but the reconstruction still understates the most extreme walls, so this must not be read as a diagnostic result.

One property of this analysis should be stated explicitly. Each patient contributes seventeen segment values, so the segment observations are not independent of one another. Segments from the same heart share the same reconstruction, the same input contours and the same anatomy. For this reason the segment values are used as a descriptive regional summary, and the group comparisons above are made at the patient level, where each patient counts once. The segment-level numbers should not be treated as if they came from independent observations.

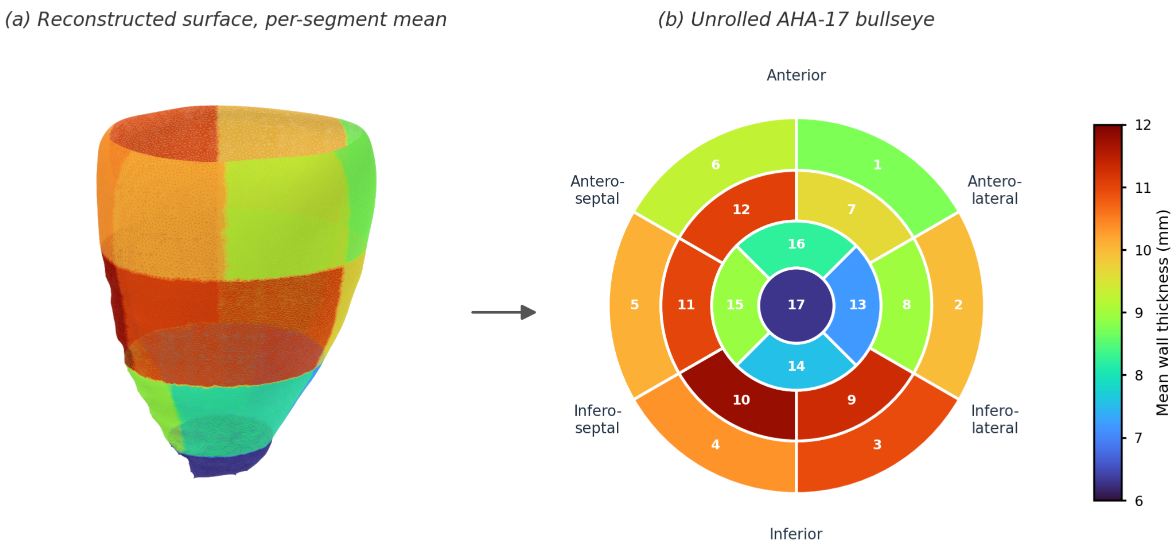

Figure 4.5. AHA 17-segment analysis of the reconstructed left-ventricular geometry at end-diastole for the representative case. Panel (a) shows the reconstructed surface partitioned into the seventeen segments, each coloured by its mean analytic wall thickness, calibrated as in Figure 4.4; panel (b) unrolls the same values onto the standard bullseye, where basal segments form the outer ring (1–6), mid-cavity segments the middle ring (7–12), apical segments the inner ring (13–16), and the apical cap the centre (17). The two panels share the colour scale, so the bullseye is a flattening of the geometry beside it rather than an independent summary.

Table 4.4. Regional wall thickness over the AHA 17-segment model for the reconstructed model geometry, measured with the Laplace field and reported separately for the 20 normal (NOR) and the 10 hypertrophic (HCM) patients. Columns give the mean end-diastolic (ED) and end-systolic (ES) thickness in millimetres and the relative thickening Δ as a percentage of the ED value. The values derive from the same sparse SAX contours as the reconstruction, so they summarise the model output and are not an independent reference.

		Normal (NOR, $n = 20$)			Hypertrophic (HCM, $n = 10$)		
#	Segment	ED	ES	Δ (%)	ED	ES	Δ (%)
1	Basal Anterior	8.72	12.25	+40.4	12.54	16.62	+32.6
2	Basal Anteroseptal	8.81	12.10	+37.4	12.83	16.02	+24.9
3	Basal Inferoseptal	8.68	11.96	+37.8	12.32	16.35	+32.8
4	Basal Inferior	8.16	12.53	+53.6	10.96	16.37	+49.4
5	Basal Inferolateral	7.94	12.81	+61.3	10.91	17.19	+57.5
6	Basal Anterolateral	8.38	12.12	+44.6	11.83	17.37	+46.9
7	Mid Anterior	7.94	12.10	+52.5	11.70	17.16	+46.6
8	Mid Anteroseptal	8.55	12.98	+51.7	13.99	18.10	+29.4
9	Mid Inferoseptal	8.57	12.98	+51.4	13.43	19.17	+42.8
10	Mid Inferior	7.93	12.46	+57.1	11.89	18.32	+54.1
11	Mid Inferolateral	7.96	13.02	+63.5	11.35	18.84	+66.0
12	Mid Anterolateral	7.92	12.39	+56.4	11.13	17.73	+59.3
13	Apical Anterior	7.28	11.75	+61.4	10.96	17.40	+58.7
14	Apical Septal	7.49	12.00	+60.2	11.33	17.59	+55.3
15	Apical Inferior	7.28	11.72	+61.1	10.76	16.43	+52.6
16	Apical Lateral	7.33	12.03	+64.0	10.61	17.28	+62.9
17	Apex	6.80	10.32	+51.8	9.67	14.71	+52.1
Overall		7.99	12.21	+52.9	11.66	17.23	+47.8

4.4. Discussion

The evaluation supports one central design choice of the model and qualifies another. The coupled endocardium–epicardium parameterisation does what it was introduced to do at the level of the field: the predicted wall offset is positive everywhere, so a thickness value exists at every myocardial point and never collapses to zero. On that geometry the three field-based estimators return wall thicknesses within about one millimetre of each other, between 9.18 and 10.01 mm. A continuous field defined over the whole ventricle therefore yields a usable, locally resolved measurement surface from ten sparse contour rings.

The watertightness that motivated the signed-distance formulation is also achieved: every endocardial and epicardial reconstruction in the cohort closes into a manifold, and the volume ratios are 0.92 for the cavity and 1.00 for the epicardial solid, so the outer shell is recovered without systematic bias while the cavity is consistently about 8% too small. Surface distances alone would not have revealed that residual shrinkage, which is why the volumetric ratios are reported beside them. This outcome still depends on the explicit repair stage described in Chapter 3: the formulation delivers surfaces that can be closed reliably, not surfaces that are closed by construction, and the thesis should not be read as claiming otherwise.

The comparison between the two clinical groups is the more informative result. The model does not collapse abnormal ventricles towards a normal mean shape: the hypertrophic group is reconstructed 3.67 mm thicker than the normal group at end-diastole, and the excess is concentrated in the septal and anteroseptal segments, which is where hypertrophic cardiomyopathy is expected to appear. The reconstruction quality itself degrades only mildly on those cases, from a 1.12 mm to a 1.43 mm endocardial Chamfer distance. The limitation is one of degree rather than of direction: taking the thickest segment of each patient as a severity index, only three of the ten hypertrophic cases exceed the 15 mm clinical threshold, so the most extreme walls are still under-stated even though the group separation is clear.

The end-systolic phase remains the harder of the two. The normal group thickens by 52.9% between the two phases and the hypertrophic group by 47.8%, and in both the thickening is stronger in the apical segments than at the base. Two properties of the design plausibly contribute. The phase information reaching the model is a single binary label rather than a continuous position in the cardiac cycle, and the synthetic pre-training corpus contains end-diastolic shapes only, so the learned prior leans towards the diastolic configuration. Because the epicardium is not represented directly but as an offset from the endocardial level set, it also accumulates the error of both the endocardial field and the offset head.

This particular limitation now has a direct remedy. The shape-model repository used for pre-training has since been extended with an end-systolic model, which was not available while the experiments reported here were being run. The synthetic corpus could therefore be rebuilt for both phases instead of one, so that the end-systolic cases are pre-trained on a prior that matches their own geometry. This is the most promising single change for reducing the gap between the two phases, and it would require no modification to the architecture or to the loss.

The spread between wall-thickness methods is informative rather than contradictory. On the reconstructed geometry the three field-based estimators agree on the general magnitude of the wall, between 9.18 and 10.01 mm, and reproduce one another point by point with intraclass correlations of 0.77, whereas the ray-based estimator reads 12.47 mm and reaches only ICC = 0.28 with limits of agreement of $\pm$12.74 mm. Field-based and ray-based estimators follow different paths through the wall, and the difference between those paths grows where the wall curves, which is where a single ray leaves the myocardium early or late. No estimator is treated as a ground truth here, and the Laplace field is used for the regional analysis because it is the established transmural-path method and its streamlines remain inside the wall by construction, not because it is closer to true anatomy.

4.5. Threats to Validity

The most important limitation concerns the reference. No public dataset provides independently acquired 3D ground-truth surfaces for these scans, so the reconstruction is evaluated against geometry *derived* from the segmentations. The Chamfer distances therefore measure agreement with a derived target rather than with true anatomy. The same caveat applies to the wall-thickness comparison: the segmentation is a stack of contours separated by several millimetres, so no source used in this chapter carries genuine through-plane information, and the way that gap is interpolated shifts the measured thickness. Absolute millimetre values should therefore not be over-interpreted; the comparisons that the data support are relative ones between pipelines evaluated on identical input. For the same reason, the wall-thickness study is an inter-method and reproducibility exercise on shared geometry, not a clinical validation. Without expert manual wall-thickness measurements, this work can claim an objective and locally resolved computational framework, but not the replacement of clinical measurement protocols.

Several further factors bound the external validity of the results, and the whole evaluation should be read as preliminary. The cohort holds 30 patients, 20 normal and 10 with hypertrophic cardiomyopathy, all reconstructed at end-diastole for the geometric metrics. This is enough to characterise the behaviour of the pipeline but far too small to support claims of clinical or pathological generalisation, and the hypertrophic subgroup in particular is only large enough for a directional observation. The regional analysis carries a further dependency: each patient contributes seventeen segments, so the segment values are repeated measures within a patient rather than independent samples, and they are reported as a descriptive summary for that reason. The relative thickening figures deserve particular caution, because dividing a systolic by a diastolic thickness amplifies small differences in where the wall is sampled. The wall-thickness statistics also depend on the AHA orientation and on the inference grid, both of which introduce discretisation effects that a finer grid or an alternative anatomical frame could shift, and the thickness values shown in the figures are calibrated to the segmentation reference, so their absolute level is inherited rather than predicted. The reconstruction additionally inherits the quality of the input segmentation, and the training data span more than one collection, so some domain shift between acquisition sources is expected. Finally, the synthetic epicardium used during pre-training is a geometric construction rather than a measured surface, which the fine-tuning stage on real data is designed to correct but cannot fully rule out as a residual prior. A larger cohort, and ideally a dataset with genuine 3D ground truth, would be needed to confirm the results reported here.

CHAPTER 5

Conclusions

This chapter summarises the contributions, revisits the research questions, and outlines directions for future work.

5.1. Summary

5.2. Revisiting the Research Questions

5.3. Contributions

5.4. Limitations

5.5. Future Work

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APPENDIX A

Supplementary Methodological Details

This appendix records supporting methodological material that would otherwise repeat information established in the main text.

A.1. Training data streams

All three training streams use the shared cache representation described in Section 3.2.5. Table A.1 provides a compact reference for their source, cardiac phase, and role in the two-stage training procedure, complementing the source datasets listed in Table 3.2. Figure A.1 presents the same composition graphically.

Table A.1. Data streams sharing the cache schema and their role in training.

Stream	Phase	Source	Training role
Synthetic SSM	ED	UK Digital Heart Project LV SSM	Pre-training to establish a geometric shape prior
Real MRI	ED	ACDC and original M&Ms	Fine-tuning on patient-specific diastolic anatomy
Real MRI	ES	ACDC and original M&Ms	Fine-tuning on systolic anatomy with phase conditioning

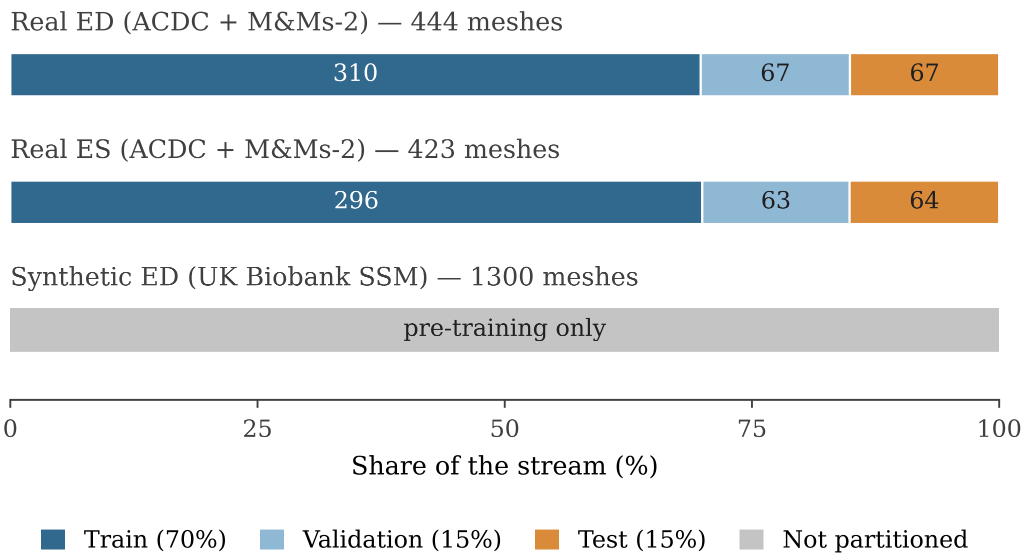

Figure A.1. Composition of the three data streams. Each bar shows how the valid samples of one stream are distributed over the 70 %/15 %/15 % train/validation/test partitions, with the number of patients annotated inside each segment. The synthetic stream, of which 1300 meshes were accepted out of 2048 sampled shape vectors, is reserved entirely for pre-training and is therefore not partitioned.

A.2. Real ED and ES mesh-preparation pipeline

Table A.2 lists, in order, the operations that convert a clinical segmentation stack into the watertight mesh used as derived supervision. The four stages of this conversion are described in Section 3.2.4.

Table A.2. Main operations in the real ED/ES mesh-preparation pipeline.

Operation	Purpose
RAS+ canonicalisation	Standardises patient-coordinate orientation while preserving physical spacing
Label harmonisation	Converts ACDC and original M&Ms to a common LV/myocardium mask
Isotropic resampling at 1.5 mm	Reduces anisotropy before 3D surface extraction
Per-slice cleaning	Removes small 2D fragments and fills holes
3D largest-component filtering	Keeps the main anatomical LV region
Gaussian mask smoothing and marching cubes	Suppresses staircase artefacts and extracts the surface
Taubin smoothing and Laplacian fairing	Reduces residual artefacts while preserving global shape
Basal cap perpendicular to the apex–base axis	Replaces the non-anatomical basal dome at the valve-plane boundary
Quadric decimation to 4000 vertices	Produces a consistently sized mesh for all cases
Plausibility checks (7 criteria)	Rejects geometrically implausible or corrupted cases